UNIT IV

UBIQUITOUS COMMUNICATION

CONTENTS

- Audio Networks
- Data Networks
- Wireless Data Networks
- Universal and Transparent Audio, Video and Alphanumeric Data Network Access
- Ubiquitous Networks
- Network Design Issues

- Ubiquitous applications use communication networks to access relevant remote external information and tasks, anywhere and anytime. Although, communication access can be modelled as part of the internal system, the core of the communication network infrastructure is considered to be external to the UbiCom system and part of the system's virtual computing environment..

- Different applications require different combinations of network functions and services, e.g., data streaming, minimal jitter, type of media access control, etc. Different networks support different sets of communication functions in different ways.

- Key design issues concern, first, whether or not these communication functions are largely transparent to services or whether or not communication is exposed via some interfaces and able to be configured and controlled by services .

- Second, there is the issue of whether or not to make all networked services ubiquitous, attached anywhere and accessible from anywhere just in case these services may be needed, versus selectively accessing networked services, e.g., some services may be limited to a locality.

AUDIO NETWORKS

Audio networks were the first types of pervasive communications Networks.

There are two basic types: audio unicast networks (PSTN) and an audio broadcast (radio) networks.

- PSTN Voice Networks
- Intelligent Networks and IP Multimedia Subsystems
- ADLS Broadband
- Wireless Telecoms Networks
- Audio Broadcast (Radio Entertainment) Networks

➢ PSTN Voice Networks

- Networks based upon Public Switched Telephone Network (PSTN) are designed to support voice communication. This originally used analogue transmissions but this has been replaced by digital transmission in the core network as this is more cost effective.

- Telephones act as access devices to the PSTN and are typically connected to a private circuit switched network or Private Branch Exchange (PBX) at work.

- The PBX controls access to a smaller number of external connections. Individual home users tend to be connected to a PSTN using a single line local loop to a local switching station.

- The PSTN can also be accessed using fixed or mobile phones. Many users also use the PSTN as an access network to data exchange networks, i.e., the Internet.

- Computer systems at work tend to connect to separate Local Area Networks, and then use a router to connect to an external network managed by an Internet Service Provider (ISP).

➢ Intelligent Networks and IP Multimedia Subsystems

- The Intelligent Networks (IN) network service model, standardised by the ITU T enables Telcos to offer new value added and customised voice services such as toll free calls.

- This supports independent component based services in general purpose computer nodes rather in special switching nodes.

- This enables service providers to drive new services rather than network providers and allows them to use these to form flexible overlay networks.

- The emphasis is now more on the development of telecom services and APIs rather than on developing new telecom network protocols.

- Although there seems to be a clear move to IP based networks, in the shorter term, hybrid IN and Internet service architectures for mobile users are being proposed such as IP Multimedia Subsystems (IMS).

➢ ADLS Broadband

- Asynchronous Digital Subscriber Line (ADSL) transmission technology can be used to increase the transmission capability over existing physical, e.g., copper wire PSTN type, access networks.
- It does this by exploiting the fact that audio telephony signals require only about a 3 kHz bandwidth but a typical line can transmit usable signals up to approximately 1MHz.
- The signal to noise ratio for transmissions is more dependent on distance. However, Digital Signal Processing (DSP) can be used to support signal modulation, commonly based upon Discrete Multi-Tone (DMT) an international standard, and Carrierless Amplitude modulation Phase modulation (CAP) to encode signals for improved transmission and then to recover the original signals.
- At both the exchange and the customer premises, devices called splitters separate/combine the existing telephony signals from the ADSL signal. Higher transmission rate services such as a higher speed Internet access and video on demand are also available depending on the performance characteristics of the underlying physical network.

➢ Wireless Telecoms Networks

- To support access anywhere for mobile or cell phone users, wide area wireless telecoms networks have been established.

- There are different networks depending on geographic region and on the Generation (G) of the wireless network such as 1G analogue and 2G digital.

- These differ primarily in the way they are designed to share access to the wireless network among different users.

- Global System for Mobile Communications (GSM) is a 2G, Time Division Multiple Access (TDMA) network prevalent in Europe, parts of Asia, Africa and Australia but not in the USA and the Far East. Code Division Multiple Access (CDMA) is another 2G digital cellular network system mainly used for cell phones in North America.

- These networks can interoperate via gateways to allow CDMA phones to call GSM phones but a CDMA phone can’t be used directly on GSM network and vice versa.

- **DECT** (Digital Enhanced Cordless Telecommunications) is has been deployed in over 100 countries worldwide to access wireless voice communication within a local area.
- It uses both TDMA and Time Division Duplex algorithms to avoid interference from other DECT system.
- DECT also supports data exchange using the DECT Packet Radio Service (DPRS) at data rates of up to 2Mbit/s using demodulation and it supports Multimedia Access using the DECT Multimedia Access Profile (DMAP).
- A low cost digital system based upon DECT, called corDECT, has also been used to provide wireless voice and data services in rural areas.
- In terms of applications and market, DECT and Bluetooth are similar. , the majority of WAN mobile phones, games consoles, etc. use Bluetooth.
- Hence Bluetooth rather than DECT seems to becoming more pervasive for a greater range of devices.
- Although DECT seems well established to access wireless telecoms networks services in many countries, it is also faces strong competition here from WAN mobile phones.

➢ Audio Broadcast (Radio Entertainment) Networks

- There are several benefits in using audio broadcasting or radio. It supports one sender to many receivers.

- It is more ubiquitous than video as it supports a multi modal interface that allows humans to listen to music and voice while engaged in many everyday physical world activities that require visual concentration, e.g., driving a vehicle.

- Third, radio (and TV) receivers, unlike wireless data networks, are inherently designed to handle and tune into receive a wide spectrum of RF broadcasts on multiple channels.

- Digital radio has been introduced as a replacement for analogue radio.

- For digital radio, Digital Audio Broadcast (DAB) standard is most commonly used and is coordinated by the World DMB Forum.

- DAB uses the MPEG 1 Audio Layer 2 audio (MP2) codec for audio broadcasting while personal players use the MP3 codec.

The main original objectives of DAB were:

1. to provide radio at CD quality
2. to provide better in car reception quality than using FM analogue radio
3. to use the spectrum more efficiently
4. to allow tuning by the name of the station rather than by frequency
5. to allow data to be transmitted.

Data Networks

Data Networks

The early Internet in the1960s was based upon several innovations.

- First, it was based upon the shift from (single tasking) batch computers to (multitasking) time shared computers.

- Second, the original intention was to directly connect each computer node to each other in the form of a peer to peer (P2P) mesh network.

- Third, a shift from analogue to digital communication was needed in order to avoid signal degradation across multi node networks.

- Fourth, a network is needed to support high capacity and resilient network paths.

- Fifth, large data was split into fixed size data packets and, sixth, there was a shift from a circuit switched Telecoms network model to a packet switched data model.

It includes

- Network Protocol Suites
- Addressing
- Routing and Internetworking

Network Protocol Suites

- Network protocols generally define a fixed length for pieces of data to be transmitted, called the data packet size.
- This makes it easier to store and forward data packets when sending and receiving and to transmit data more reliably and efficiency.
- Data messages that exceed the fixed length of the data packet size can be split into smaller related individual data packets. This is called data segmentation. To send and receive content data packets, additional control packets are used such as synchronisation and acknowledge data packets.
- The types of data and control packets are defined in a network communication protocol. Each data packet is labelled with the address of an end receiver computer node in a particular network.
- This enables packets from multiple messages to be interleaved or multiplexed to use the same part of the network. Each type of physical network link protocol defines its own type of data packet and multiplexing.
- Data packets defined by one data protocol can be tunnelled or encapsulated in other types of data packet.

Addressing

- Before communication can occur between network elements such as computers, they need to be allocated network addresses.
- In computer networks, network addresses are allocated to networks in logical ranges called subnets.
- To transfer messages across networks requires the use of a special network computer called a router.
- Network addresses consist of two parts: a (sub) network address and a host computer address.
- IPv4 supports 32 bit or 232 addresses. IPv6, however, supports 128 bit or 2128 addresses.

Routing and Internetworking

- Multiple paths may be available between sender and receiver nodes located on different networks which may be interlinked to form an Internet.

- Data may be too large to be transmitted in one go and may need to be split into multiple packets, that can get transmitted along multiple routes, referred to as data routing.

- Routers communicate with each other using specialised routing protocols that provide updates to the routing tables.

- One of the most common used routing algorithms is Open Shortest Path First (OSPF). Each OSPF router broadcasts its routes and in turn requests information about routes that it can connect to.

Wireless Data Networks

There are several benefits to using wireless networks for UbiCom:

- ***Anywhere***: In contrast to wired networks which can only be accessed at a fixed number of network junctions, wireless networks give users the freedom to access them anywhere where they are still in range of a wireless transmitter or hub that they have access permission for.

- ***Mobility***: Wireless communication networks can be accessed while moving. The cost of installing wireless transmitters and receivers typically is much cheaper than a wired network.

- ***Less disruptive***: Wireless networks can be used in areas where wired networks would be con sidered too inconvenient, disruptive or expensive to install, e.g., in old historical buildings and in emergency situations.

- ***Adaptivity***: Wireless networks are also considered more adaptive in terms of their ability to expand or shrink the coverage of the network and to vary the density of coverage, installing more transmitters and capacity in high populated in contrast to more rural areas.

Types of Wireless Network

- Networks vary according to the type of infrastructure, the network range, frequencies used, the type of signal modulation to increase channel efficiency and channel sharing, the bandwidth available and power consumption.

- For example, most global, wide area and local area wireless networks are infrastructure dependent and use fixed transmitters, e.g., satellite, mobile phone, WLAN or WiFi, WiMax.

- In contrast, in an ad hoc wireless network the transmitters and routers are dynamic, e.g., packet radios and sensor nets.

- Mobile wireless networks can vary in the range they cover. They can be global, e.g., satellite; wide area covering 100s to 1000s of km, e.g., mobile phone networks such as GSM, TDMA, CDMA; local area and metropolitan networks covering 100m to 5 km, e.g. WLAN; or personal networks covering 1 10 m, e.g., Bluetooth, ZigBee and InfraRed.

WLAN and WiMAX

- Wireless LANs, also called WLANs are local area wireless networks that adhere to the IEEE 802.1l set of standards that can operate at different frequency ranges and support different message transfer rates.

- This works in the 2.4 GHz band (like 802.11b) and operates to a maximum raw data rate of 54 Mbps.

- A typical WLAN network consists of computers with WLAN cards that connect to WLAN access nodes that have a wired connection to an internet.

- WLAN currently tends to be inbuilt into laptop computers but not currently in many mobile phones.

- Wi Fi is a registered trademark of the Wi Fi Alliance, a trade organisation that tests and certifies equipment compliance with the 802.11x standards.

- **WiMAX**, the Worldwide Interoperability for Microwave Access, from the WiMAX Forum , is proposed as wireless wide area broadband access technology, based upon the IEEE 802.16 standard, typically offering 10 Mb/s over 10 KM although speeds up to 70 Mb/s are achievable over 10 KM.

- The IEEE 802.11 and 802.16 technologies are distinguished by their type of medium access control (MAC) whereas 802.11 MAC uses CSMA (‘listen before talk’) and is connectionless.

- The 802.16 MAC supports full QoS, bandwidth on demand, is connection oriented; it supports centralised control and scheduling and offers multimedia support.

Bluetooth

- Bluetooth is a standard for short range wireless communication over about 1 100 m depending on the class of device and power.

- Bluetooth applications include both local communication and increasingly local control.

- Unlike IR, Bluetooth does not require a line of sight between the transmitter and receiver.

- Some current popular Bluetooth devices and applications include hands free mobile phone headsets and car kits to support hands free communication when driving and wireless controllers of game consoles.

- Bluetooth uses the same radio frequencies as WLAN but with higher power consumption resulting in a stronger connection.

ZigBee

- ZigBee is a specification for a suite of communication protocols from the ZigBee alliance.
- It uses small, low power digital radios based on the IEEE 802.15.4 standard for Wireless Personal Area Networks (WPAN) as a beaconing technique in which a node continuously transmits small packets to advertise its presence to other mobile units.
- This then tries to establish a connection to start networks, then letting other devices join in. ZigBee operates in the industrial, scientific and medical (ISM) radio bands; 868 MHz in Europe, 915 MHz in the USA and 2.4 GHz in most jurisdictions worldwide.
- ZigBee protocols are intended for use in embedded applications requiring low data rates and low power consumption and low latency.
- ZigBee's current focus is to define a general purpose, inexpensive, self organising, mesh network that can be used for industrial control, embedded sensing, medical data collection, smoke and intruder warning, building automation and home automation.

- There are three different types of ZigBee device.

➢ **A ZigBee coordinator (ZC)** forms the root of the network tree and can bridge to other networks.

➢ **A ZigBee Router (ZR)** acts as an intermediate router, passing data from other devices.

➢ **A ZigBee End Device (ZED)** contains just enough functionality to talk to its parent node (either the coordinator or a router) It cannot relay data from other devices and requires the least amount of memory.

- ZigBee protocols use an ad hoc, on demand, distance vector to automatically construct a low speed ad hoc network of nodes.

- In most large network instances, the network will be a cluster of clusters. It can also form a mesh or a single cluster.

Infrared

- Infrared (IR) is a short range low bandwidth data communication used to communicate between computers and between computer peripherals.

- Remote controls and IrDA devices use infrared light emitting diodes (LEDs) to emit infrared radiation which is focused by a plastic lens into a narrow beam.

- The beam is modulated, i.e. switched on and off, to encode the data. The receiver uses a silicon photodiode to convert the infrared radiation to an electric current.

- It responds only to the rapidly pulsing signal created by the transmitter, and filters out slowly changing infrared radiation from ambient light. Infrared communication is more useful for indoor use.

- IR requires a line of sight from the transmitter to the receiver. It does not penetrate walls and so does not interfere with other devices in adjoining rooms.

- Infrared is the most common way for remote controls to command appliances.

UWB

- Ultra Wideband (UWB) is a technology for transmitting information at data rates exceeding 100 m bits/s, spread over a large bandwidth (>500 MHz) at a low power range, over short distances.
- This is intended to provide an efficient use of scarce radio bandwidth while enabling both high data rate wireless connectivity within BANs, PANs and within buildings and at longer range, low data rate applications, as well as radar, collision obstacle avoidance, precision altimetry and imaging systems .
- In contrast to conventional wireless systems that need to use baseband signals to modulate radio frequency (RF) carrier signals, UWB can be used to directly transmit signals at baseband frequencies.
- In pulse based UWB, the transmitter only needs to operate during the pulse transmission, producing a strong duty cycle on the radio and minimising baseline power consumption.
- As most of the complexity of UWB communication is in the receiver, simple, low power transmitters can be supported.

Satellite and Microwave Communication

- Satellite communication has the potential for truly ubiquitous global communication. Commercial satellites use parts of the microwave range frequencies for transmission.
- First communication satellites in the early , did not provide global coverage but acted as mobile relays to store and forward transmissions between ground stations in different locations, hence ground stations could only communicate for part of the day.
- In the mid 1960s, so called geostationary satellites, which remain in a fixed location and operate in the 4 and 6 GHz range, the C Band, were launched.
- Geostationary satellites have the advantage of much simpler antennae design and configuration and a relatively small number of these satellites can be interlinked to provide global coverage.
- Satellite design must contend with station keeping design, e.g., the influence of the sun and moon in causing orbit perturbations, communication payload design, handling the large round trip delay for transmitting and receiving signals and efficient multiple channel utilisation.

Roaming between Local Wireless LANs

- There are several reasons for using Internet wireless networks:

➢ To support roaming between short range wireless networks,

➢ To select an optimum network when several overlap, e.g., based upon bandwidth, security etc, and

➢ To allow short range services to interconnect with longer range ones, e.g., to notify someone remotely that something of interest remote is happening locally.

- For example, logistics companies may use WLAN access points in warehouses to interlink scanners that audit tagged goods that are being loaded, unloaded and being stored.
- Global positioning system capabilities built into the device can help dispatchers know the exact location of any truck and warn a driver about to deliver a package to the wrong location.

- Generic Access Network (GAN), also known as Unlicensed Mobile Access (UMA), is a telecommunication system allowing seamless roaming and handover between local area networks and wide area networks using dual mode mobile phones.
- The aim of GAN is to enable GSM mobile operators to offer fully converged connectivity using their existing core network.
- Subscribers could then seamlessly roam from one cellular network to a WLAN, maintaining the call as they move from one to the other.
- Although GAN technology is already available and deployed by a number of cellular carriers, it is not yet used in connection with 3G wireless telecoms networks.
- In contrast, Femtocells are small cellular access points which provide enhanced coverage specifically in residential environments, enabling operators to provide fixed mobile converged voice, data and video services such as IPTV

Universal and Transparent Audio, Video and Alphanumeric Data Network Access

- Different content media are delivered over different types of network because they needed to use networks with different properties and because they were developed by separate technical groups.

- Audio and Video (AV) broadcast Content Based Networks (CBN) have different drivers compared to telecoms and network networks. The first obvious difference is that digital AV CBN transmits streamed multimedia audio and video.

- Second, broadcast networks are designed more for simplex or one way, one to many, synchronised transmissions. In contrast, telecoms networks are developed to support duplex or two way, one to one communication. Whereas the Internet was developed, at least initially, to support asynchronous communication.

- Combined Voice and Data Networks
- Combined Audio-Video and Data Content Distribution Networks
- On-demand, Interactive and Distributed Content

Ubiquitous Networks

Wireless Networks

UNIT III

Smart Environments

Smart Environments

- Tagging, Sensing and Controlling,
- Tagging the Physical World,
- Sensors and Sensor Networks,
- Micro Actuation and Sensing: MEMS,
- Embedded Systems and Real Time Systems,
- Control Systems.

Tagging, Sensing and Controlling

- As electronic components become smaller, faster, and cheaper to fabricate, more low power and more low maintenance.

- MEMS form a powerful enabler for the vision of smart ubiquitous computing environments. Micro Electro Mechanical Systems (MEMS) will enable sensing and actuation down to a scale of a nanometre.

- Embedded systems are a second IT trend to enhance natural or artificial physical objects with digital, networked embedded devices to provide improved manual, semiautomatic and automatic sensing and control, e.g., in motor vehicles.

- A third trend is the increasing annotation of the physical world, often driven by a need to enrich human and other natural interactions in these environments, to be better informed about these interactions at both a local and global level. For example, we may want to locate where we have left things such as our keys or phones, or we may want to learn where an insect we have never seen before, originates from.

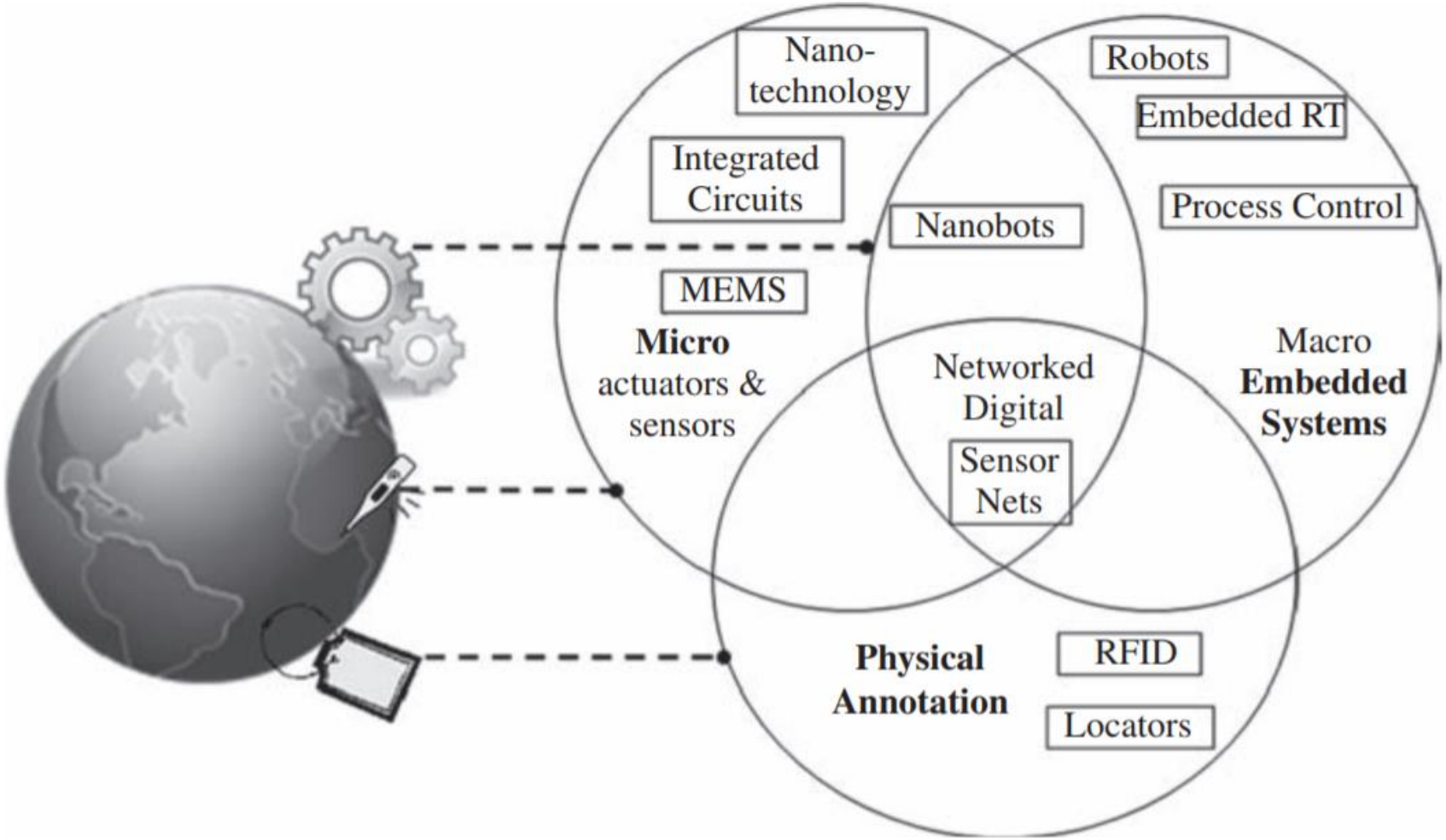


Figure 6.1 Enabling ubiquitous computing via micro, macro embedded and annotation of physical objects in the world

Three complementary enabling trends for pervasive computing are described.

- First, tagging of objects in the physical world, using a mixture of sensors and tags, enables these features to form part of a discourse in a virtual computer environment.

- Second, mechanisms can be reduced to millimetre, micrometre and nanometre sizes to be more easily blended into the physical world. Design issues in creating and operating much smaller devices are considered from an ICT engineering point of view.

- Third, digitally enabling and networking macro mechanisms and devices are accompanied by embedded process control computer systems including robots to automate interaction with humans and to allow systems to operate autonomously.

Tagging the Physical World

Tagging the Physical World

- Physical tags refer to digital tags, which are networked electronic devices with an identity, e.g., a RFID (Radio Frequency Identifier) tag.

- When these are attached to or linked to physical objects, they provide a way to audit physical spaces and processes. An important motivation for the use of physical world tags is to support context based querying and tracking of physical world objects.

- Tagging is also an enabler for augmented reality (AR) which deals with mapping physical world objects into computer artefact objects in order to more conveniently access them and to manipulate them in the virtual environment.

Life-Cycle for Tagging Physical Objects

- **Capturing** a physical view or recording of physical objects or some object feature.
- **Identifying** physical objects.
- **Anchoring** or relating objects.
- **Organisation or structuring.**
- **Presentation**
- **Management**

Tags: Types and Characteristics

- Augment the user by generating a digitised view of part of the physical world while the user is onsite in that part, e.g., use AR.

- Augment the physical object by directly attaching to, or embedding a physical (digital) tag into, the physical object, e.g., use RFIDs.

- Augment the environment surrounding the user and object, e.g., digitise and present a view of the physical environment in a virtual environment while the user is collocated in that environment.

Physical and Virtual Tag Management

- Physical object annotation for business still faces many management challenges that inhibit their mass consumer use. Many of these issues are also similar for sensors too.
- Tags can be bound to objects when objects are created versus attached later.
- Tags can be permanently bound to objects versus being removable.
- Tags could be removed from objects at the end of some usage life cycle versus never removed.
- Tags can be simply disposed of when the object is disposed of.
- Tags may need to be read outdoors in noisy, wet, dark or very bright environments.
- The annotation data needs to be stored, distributed and integrated starting from the captured data; data management must start as soon as the data is captured (by tag readers).

RFID Tags

- RFID (Radio Frequency Identifier) tags can be attached to objects to enable identification of objects in the world over a wireless link.

- RFID tags may be classified into whether or not they have their own energy supply (active RFID tags) or whether or not they are passive RFID tags, using the energy supply of the reader.

- Active tags are more expensive and require more maintenance but tend to have a longer range compared to passive tags.

- A typical RFID system consists of two main components: the tag itself and a reader that scans the tag for its ID and contains additional computation, data storage and communication facilities.

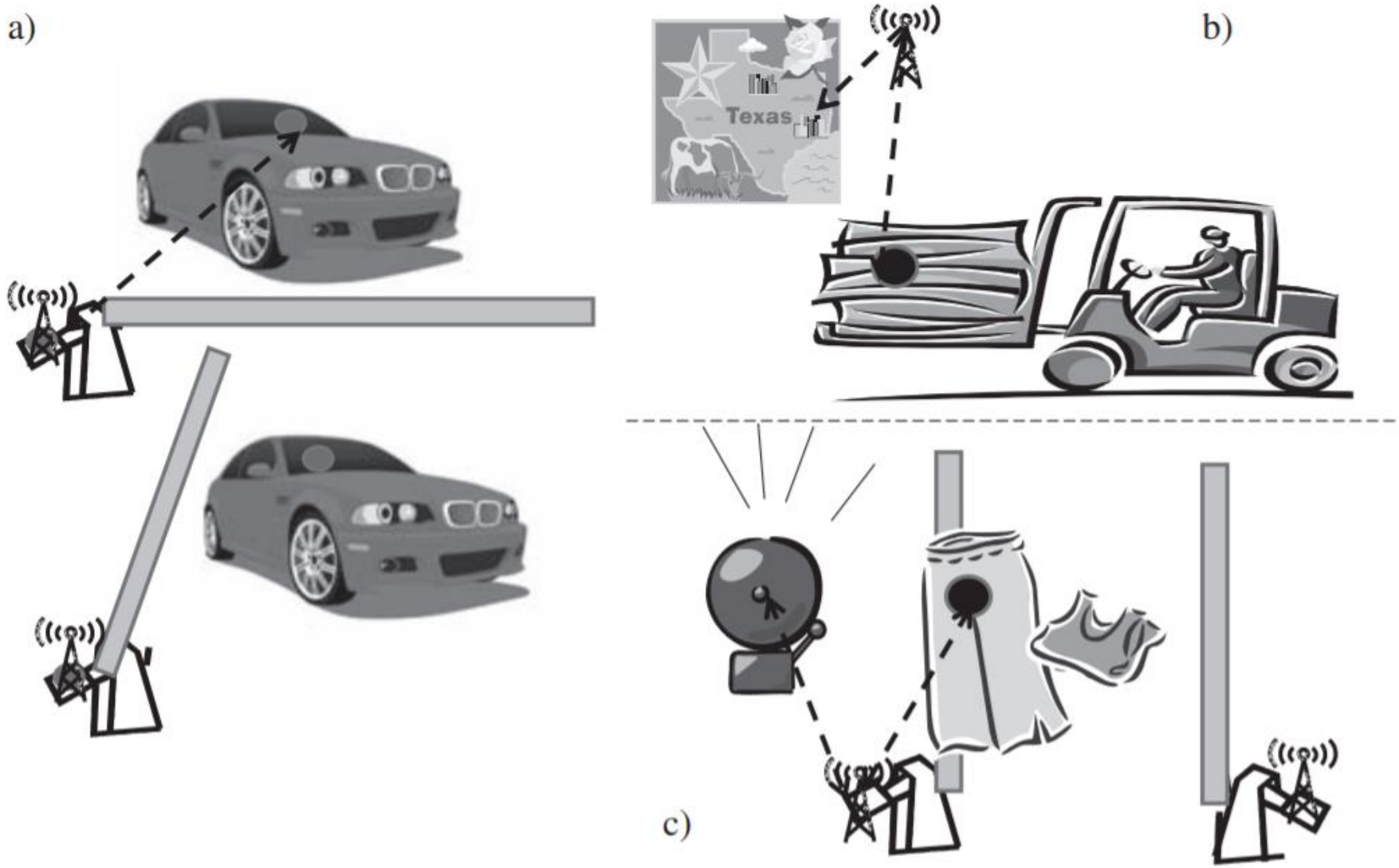


Figure 6.3 RFID tag application: (a) transponders in cars cause toll barriers to automatically lift as cars approach; (b) tags on pallet of goods tell distributers where goods are located; and (c) tags on clothes in retail outlets can signal alarms if they are removed without permission

Active RFID

- Active RFID tags are used on large assets, such as cargo containers, rail cars and large reusable containers, which need to be tracked over distances such as a distribution yard.
- There are two types of active tags: **transponders** and **beacons**.

- Active transponders are woken up when they receive a signal from a reader. An important application of active transponders is in toll payment collection and checkpoint control.

- When a car with an active transponder approaches a tollbooth, a reader at the booth sends out a signal that wakes up the transponder on the car windshield. The transponder then broadcasts its unique ID to the reader. Transponders conserve battery life by having the tag broadcast its signal only when it is within range of a reader.

- Beacons are used in a location based systems where the precise location of an asset needs to be tracked within a region such as a distribution yard or along a transport route.

- Longer range location based systems could utilise GPS or mobile phone trilateration. In a location based system, a beacon emits a signal with its unique identifier at pre set intervals (it could be every three seconds or once a day, depending on how important it is to know the location of an asset at a particular moment in time).

- The beacon's signal is picked up by at least three reader antennas positioned around the perimeter of the area where assets are being tracked.

- More complex active tags could also incorporate sensors, e.g., a tag on a logistics item could sense if it has been dropped or if its surrounding temperature has become too hot or too low.

Passive RFID Tags

- Passive RFID tags contain no power source and no active transmitter, their power to transmit their information, typically between 10 mW and 1 mW, comes from the reader.

- They are cheaper than active tags. They are lower maintenance and much shorter (read access) range than active tags, typically from a few cm to 10 m.

- A passive RFID transponder consists of a microchip attached to an antenna. Transponders can be packaged in many different ways, e.g., sandwiched between an adhesive layer and a paper label to create a printable RFID label; embedded in a plastic card, a key fob and in special packaging that can resist heat, cold or harsh cleaning chemicals

Micro Versus Macro Tags

- Most physical world tags such as RFID tags are macro sized tags to fit macro sized objects.

- RFID tags have the potential to be reduced in size based upon MEMS techniques to hundredths of a millimetre which are much cheaper to mass produce but which are invisible to the unaided human eye.

- The size of a wireless antennae transceiver for the micro tag, which can be designed to be external to the tag, is dependent on the wavelength of the wireless signal transceiver and is typically of the order of about 5 cm for a 2.45 GHz signal.

- Using MEMs and nano technology markers, we have the potential to mark up the physical world and observe and affect the world in unprecedented ways at very fine, levels of granularity, at the micro and at the nano or molecular level.

Sensors and Sensor Networks

- Sensors are a type of transducer that converts some physical phenomenon such as heat, light, sound into electrical signals.

- Sensors often act as an enabler, as inputs to a system behaviour so that it can more favourably adapt, often embedded as part of a control loop in pre programmed systems that perform specialised rather than general purpose functions, e.g., a temperature sensor may be hard wired into a heating system.

- Sensors can be used to do the following:

✓instrument and monitor environments;

✓track assets through time and space with respect to some workflow or process;

✓detect changes in the environment defined to be of significance that humans are unable or are put at risk to perceive;

✓control a system with respect to the environment within a defined range of changes; adapt services to improve their utility.

- Sensors like RFID tags are networked. The basic architecture is similar: sensors can act as data generators, intermediate or services nodes receive, post process and store data, possibly remotely.

- However, whereas tags just generate a fixed electrical signal, sensor data may change because it is the output from a transducer that converts varying physical phenomena into varying signals.

- The data processing is also likely to be more complex for sensors than tags. Sensors may range in scale from nano sensors to macro sensors such as a windsock used to indicate the wind direction.

Overview of Sensor Net Components and Processes

- The main components of a typical sensor network system are sensors connected in a network that is serviced by a sensor access node.

- A slightly different but compatible view of a sensor network is to view sensors as being of three types of node:

1. Common nodes mainly responsible for collecting sensor data;
2. Sink nodes that are responsible for receiving, storing, processing, aggregating data from common nodes; and
3. Gateway nodes that connect sink nodes to external entities

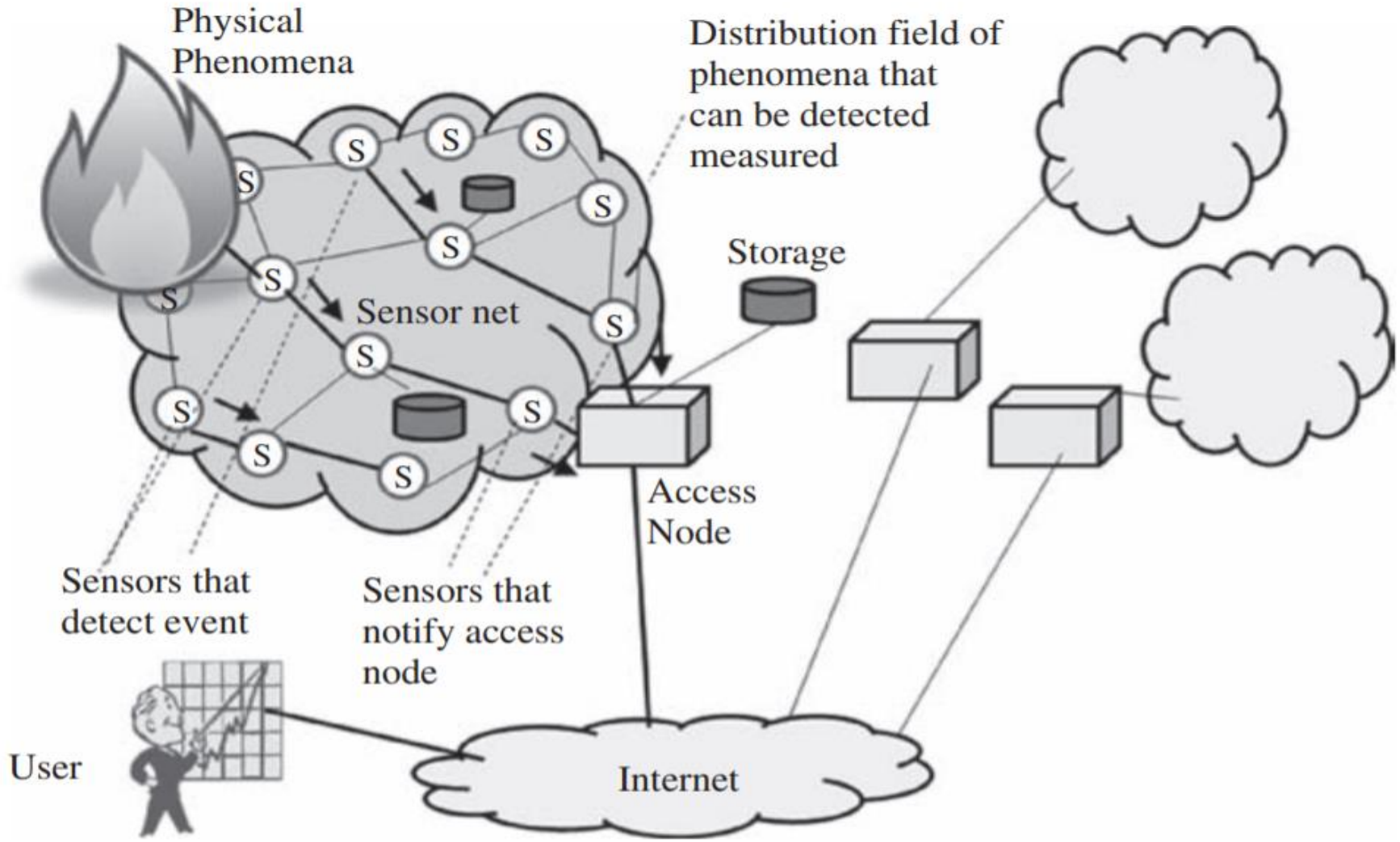


Figure 6.5 A sensor network used to detect increases in heat and report these to a user

Sensors can be characterised in terms of:

- **Characteristics of the phenomena being sensed**: The type of physical phenomenon sensed includes location, temperature, etc.

- **Variability**: of the type of environment being sensed; application tasks; the spatial distribution, i.e., objects of interest in the environment which may move.

- **Sensor physical characteristics, including power, mobility, and size**. Passive sensors transfer power from a reader whereas active sensors require their own power source. Sensors may be anchored in a fixed location or may be mobile.

- **Functional complexity**: Some sensors have no autonomy and have simple functionality. Sensors merely convert some physical phenomena into data that is simply reported to human users.

Sensor Electronics

- It split into four main functions: sensing, processing, transceiving and power related.

- The signal from the sensor is filtered and amplified, converted into a digital signal by the analogue to digital converter (ADC), some simple digital signal processing (DSP) is performed at the sensor before the signal is modulated for transmission.

- This particular sensor design also supports input configuration for the DSP.

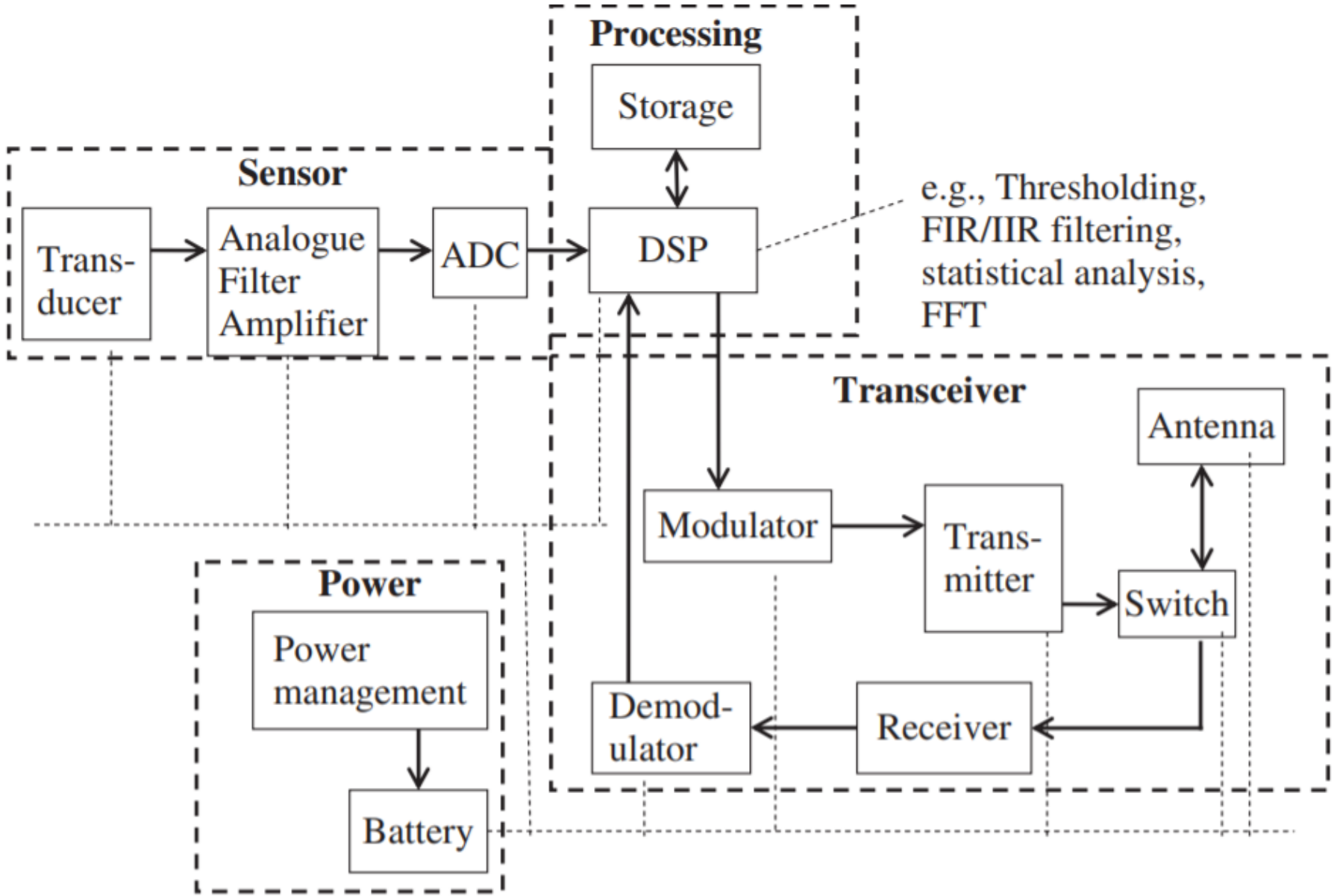


Figure 6.7 Block diagram for a sensor electronics circuit

Table 6.1 Challenges in designing and deploying sensors and some corresponding solutions

Challenges of a sensor net system	Design solutions
Sensor energy is a scarce resource for data transmission	Use a sensor net that deploys, low power, short range transmissions Network sensors into mesh networks and use multi hop transmissions Filter data in situ and transmit only filtered data
Limited memory and computation power in sensors	Harvest renewable energy from the environment and store
Dynamic and non deterministic spatial temporal distribution of events. May not be able to pre determine how to optimally deploy individual sensors	Use a sensor net to increase the sensor density around estimated signal source positions when deterministic; Design sensor distributions to be reconfigurable, self organising, to be mobile Support variable sampling and support bursty data collection
Sensor failure is common due to a lack of power, physical damage, active (jamming) or passive environmental interference of the transceiver, access node or non optimal positioning	Use dense networks of low power sensors with redundant paths to route data through the network Use of counter measures and frequency shifts Locators and trackers are needed to locate (moving) sensors and can be used to position them
Multi hop sensor networks may have a dynamic topology. No global knowledge about structure of network may be known	Use specialised routing protocols to work over dynamic mesh topologies
Sensors can be too costly to update once deployed	Design sensors and sensor access nodes to be low maintenance. Support sensor redundancy
Sensors can generate huge quantities of data	Use in situ data processing both in the sensor and the sensor access node

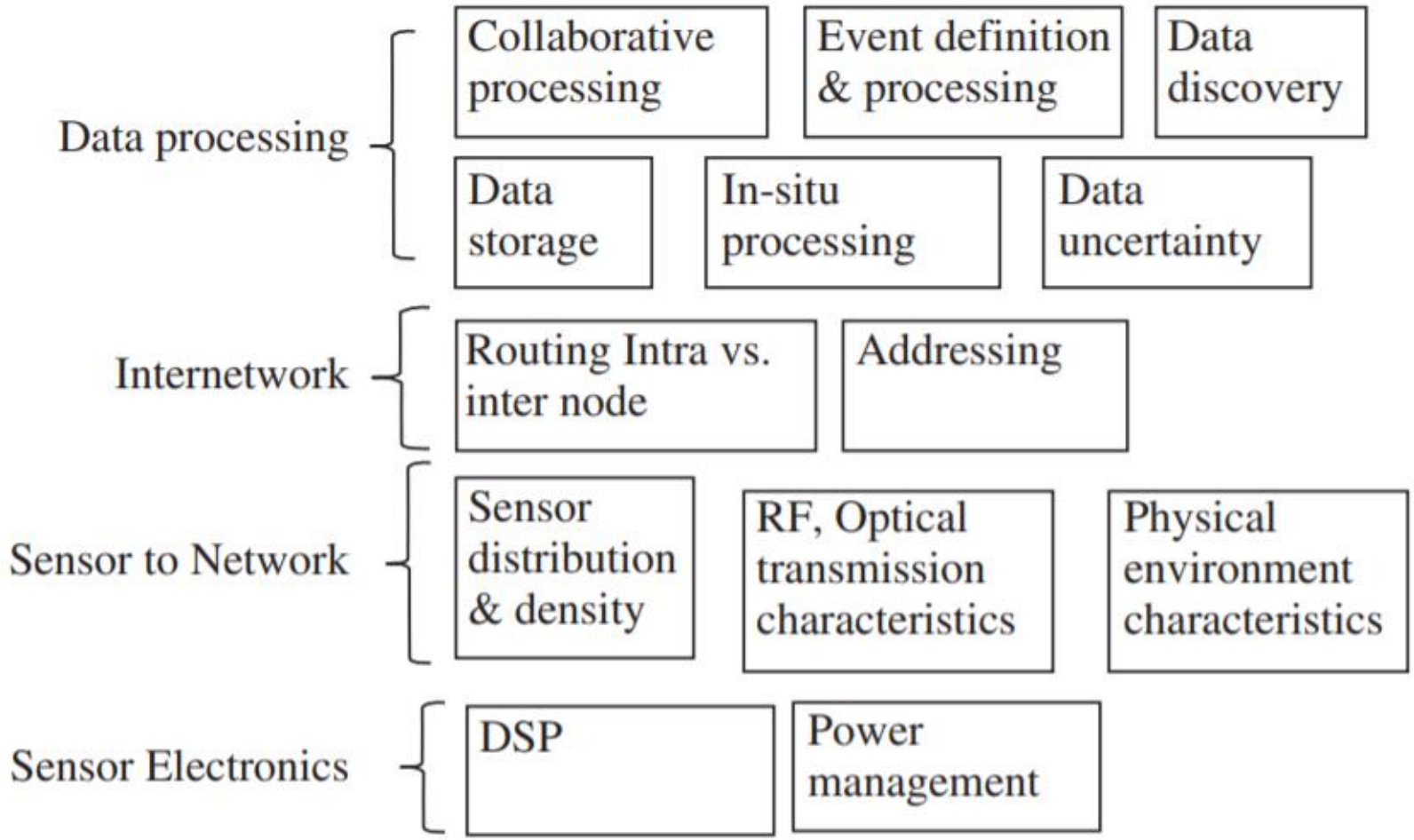


Figure 6.6 The main functional characteristics for sensor net deployment

Physical Network: Environment, Density and Transmission

Sensors may be deployed in three phases:

1) A pre deployment phase where group dispersal from multiple release and scatter points or individual placement occurs

2) A deployment phase in which sensors may move after deployment because of signal phenomena changes, energy optimisation or task changes

3) A redeployment phase where additional sensors are used.

There are several reasons motivating the use of multiple low cost, short range, low power sensors rather than using a few long range, high power and high cost sensors.

- **First**, dense networks of sensors can improve the signal to noise ratio (SNR) by reducing the average distance of sensor to signal sources of interest. Each sensor has a finite sensing range determined by the baseline (floor) noise level. A dense sensor field improves the odds of detecting a signal source within a range. Once a signal source is inside the sensing range, further increasing the sensor density decreases the average distance from sensor to signal source, thus improving the SNR.

- **Second**, the energy efficiency for communication can be increased through the use of a multi hop topology for the sensor network

Data Network: Addressing and Routing

➢Sensor Networks Versus Ad Hoc Networks

- There are several differences between sensor networks and ad hoc networks.

- In contrast to ad hoc networks, sensor networks are denser and contain several orders of magnitude more (sensor) nodes that may not be uniquely addressable at an application level because of the large number of instances of nodes of the same type.

- The topology of sensor nets can be very dynamic. The topology may need to adapt to locally detected sensors, to unknown signal characteristics, to the reconfiguration of in situ nomadic sensors and to dynamic routing, in order to optimise a low SNR and power usage.

- Sensor nodes often use a broadcast communication paradigm, whereas ad hoc networks often use point to point communications. Typically sensors use a mesh topology network , that is dynamic, mobile, and unreliable, that assumes no universal routing protocols and no central registry of sensor locations and routes. Each sensor acts a router and as a data source and depending on design also as a data sink and gateway.

Data Processing: Distributed Data Storage and Data Queries

- Sensors as data sources are naturally distributed and hence the computation becomes distributed as a consequence of performing it locally in order to reduce it.

- Processing involves collection of events from massively distributed sensors. It would be simplest to allow each sensor to act in isolation, to keep all data events but this requires a large amount of storage.

- A traditional centralised data storage approach could be used with many sensors generating data transactions that cause updates in a data warehouse. Data events extracted from sensors could be stored in a RDBMS server.

- Query processing takes place at data server nodes only. In addition, multiple data from RDBMS could be aggregated into a data warehouse system and SQL could be used to query sensor data identified using an attribute based addressing and routing scheme.

Micro Actuation and Sensing: MEMS

Micro Actuation and Sensing: MEMS

- MEMS (Micro electro mechanical systems) are micron to millimetre scale electronic devices fabricated as discrete devices or in large arrays.

- MEMS perform two basic types of functions, acting as sensors or actuators. Both actuators and sensors act as transducers converting one signal into another.

- Of specific interest are transducers that covert some environmental phenomena such as temperature, humidity, pressure, etc., into a digital electrical signal and vice versa.

- MEMS actuators convert an electrical signal into physical phenomena to move or control mechanisms such as motors, pneumatic actuators, hydraulic pistons and relays.

- MEMS sensors work in reverse to actuators, they convert some environmental phenomena such as temperature, humidity and pressure into an electrical signal. These relatively small components have high resonant frequencies leading to higher operating frequencies.

Fabrication

- MEMS design differs from that of the equivalent macro devices which comprise mechanical and discrete electronic component design because these micro components are based upon silicon based Integrated Circuit (IC), also called chip, design.

- Analogue devices may also be replaced by IC versions, e.g., whereas a traditional thermometer is based upon a liquid, such as mercury, expanding along a tube referenced to a calibrated scale, an electronic thermometer can be built out of a thermocouple and IC amplifier.

- ICs consist of several layers of p type and n type doped silicon which have been added to a substrate. An optical microfabrication approach, photolithography, is then used to fabricate the circuit.

- This **first** covers a layer with a photoresistant chemical. Then the circuit pattern to be fabricated is drawn onto a photomask.

- The photolithography systems shines the UV light through the photomask, projecting a shadow onto a layer that then reacts with the photoresistant chemical and hardens, allowing the selective removal of parts of the substrate to be chemical etched away.

- The **second** part of MEMS to supplement the microelectronics part is the micromachines part.

- These micromachines can be fabricated just like ICs.

- MEMS type ICs can be fabricated in different ways using: bulk micro machining; surface micro machining ; and by machining LIGA deep structures.

Micro-Actuators

- The mechanisms involved in micro actuation while conceptually similar to the equivalent macro mechanisms may function fundamentally differently.

- They are engineered in a fundamentally different way using integrated circuit design and nanotechnology.

MEMS actuator applications include:

- *Micro mirror* array based projectors (micro projectors) can be used to generate large screen display content from smaller devices.
- *Inkjet printers heads*: MEMS can be used to control ink deposits onto paper.
- *Optical switches*: optical cross connect switches (OXC) are devices used by telecommunications carriers to switch high speed optical signals in a fibre optic network.
- *Micro fluid pumps*: The essential components include a fluid actuator, a fluidic control device, and micro plumbing, e.g., for use in delivering medicine.
- *Miniature RF transceivers*: can replace passive low Q, where the Q factor indicates the rate of energy dissipation relative to the oscillation frequency, components in communication devices such as vibrating resonators, switches, capacitors, and inductors and put them on a single high Q MEMS RF transceiver chip.
- *Miniature storage devices*: can support gigabytes of non volatile data storage in a single IC chip, low power, and low data latency, e.g., worst case rotational latency 5 11ms, sub millisecond average access time.

Micro-Sensors

- Size reduction as paramount to make sensor nodes as inexpensive and easy to deploy as possible, e.g., to incorporate the requisite sensing, communication, and computing hardware, along with a power supply, in a volume no more than a cubic millimetre, while still achieving a suitable performance in terms of sensor functionality and communications capability.

- These millimetre scale nodes are called ‘Smart Dust’. Smart Dust can be small enough to remain suspended in air, circulated by air currents, sensing and communicating for hours or even days.

- Smart dust motes contain micro sensors, an optical receiver, passive and active optical transmitters, signal processing and control circuitry, and a thick film battery power source.

Some common MEMS sensor applications are as follows:

- ***Accelerometers*** can be used to control the safety airbag release in almost all cars today. Accelerometers detect the rapid negative accel eration of the vehicle to determine when a collision is occurring and the severity of the collision.

- ***Angular rate sensors and gyroscopes*** can be used to measure the rotational velocity or angular rate of an object. Compared to classic gyroscopes based on optical or (macro) mechanical principles, they do not need a fixed point for referencing, are very cheap and can withstand harsher environ ment.

Smart Surfaces, Skin, Paint, Matter and Dust

- MEMS can be permanently attached to some fixed substrate forming smart surfaces or be more free standing, forming smart structures that can reorganise.

- An example of a smart surface is a paint that is able to sense vibrations because it is loaded with a fine powder of a piezoelectric material called lead zirconate titanate (PZT).

- When PZT crystals are stretched or squeezed, they produce an electrical signal that is proportional to the force .

- MEMS could be mixed with a range of bulk materials, such as paints, gels, and spread on surfaces or embedded into surfaces or scattered into and carried as part of other media such as air and water.

- For example, coating bridges and buildings with smart paint could sense and report traffic, wind loads and monitor structural integrity.

- A smart paint coating on a wall could sense vibrations, monitor the premises for intruders, and cancel noise.

- Smart surfaces can also be woven out of organic polymers that have light emitting and conductive properties.

- Organic computing can be used to form smart skin and smart clothes.

- Similar to sensor nets, MEMS can also be networked in MEMS nets.

- The Smart Dust project led by Kris Pister produced prototypes of many novel types of low powered networked MEMS sensors

Embedded Systems and Real Time Systems

- Embedded systems are used mainly online for task enactment in the physical world, in contrast to general purpose computers which are often used offline for information access and sharing.

- Thus embedded computer systems differ from general purpose (MTOS) systems in three main ways:

➢ Embedded systems focus more on single task enactment.

➢ Safety criticality may be important because actions affect the physical world.

➢ Third, tasks often need to be scheduled with respect to real time constraints.

- An embedded system is programmable and contains one or more dedicated task computers, microprocessors, or microcontrollers.

- A microprocessor is an integrated circuit which forms the central processing unit for a computer or embedded controller, but requires additional support circuitry to function.

- A microcontroller is a microprocessor plus additional types of processor that supports other devices and is integrated into a single package.

- Other types of device support devices may include serial (COM), ports, parallel ports, USB ports, Ethernet ports, A/D and D/A, interval timers, watchdog timers, event counter/timers, real time clocks (RTC), Digital Signal Processing (DSP) and numeric coprocessors.

Application-Specific Operating Systems (ASOS)

- Embedded systems require an application specific operating system (ASOS) that is customisable and reconfigurable to meet the requirements of specific applications in order to provide lower cost and higher performance by eliminating general purpose MTOS features that are unnecessary for specific applications and through better tailoring those features that are included.

- A key design issue is whether or not the configuration of the MTOS is fixed at design time or can be changed during the operational lifetime to support new requirements.

- For example, an embedded environmental control system for a smart home uses a microcontroller executing an ASOS which does not require many features of the file system, inter process communication or control, networking and security facilities provided by a MTOS.

Real-Time Operating Systems for Embedded Systems

- Real time embedded systems applications are a subset of embedded system applications which perform safety critical tasks with respect to time constraints because if these are violated, the system may become unsafe, such as the operation of cars, ships, airplanes, medical instrumentation monitoring and control, multimedia streaming, factory automation, financial transaction processing and video games machines.

- Real Time Operating Systems or RTOS can be considered to be a resource constrained system where the primary resources, such as data transfer, are constrained in time. This in turn constrains the number of CPU cycles but other resources may also be constrained such as memory.

- A RTOS reacts to external events that interrupt it.

- Timing constraints can vary between different real time systems. Therefore, an RTS can fall into one of three categories: **soft, hard or firm**.

- In a **soft** RTS, the timing requirements are defined by using an average response time. A single computation arriving late is not significant to the operation of the system, though many late arrivals might be. An example of this is airline reservation systems. If a single computation is late, the system's response time may lag. However, the only consequence would be a frustrated potential passenger.

- In a **hard** RTS, the timing requirements are vital. A response that's late is incorrect and system failure will result. Activities must be completed by a specified deadline, always because if a deadline is missed, the task fails. Deadlines can be a specific time, a time interval, or the arrival of an event. This demands that the system has the ability to predict how long computations will take in advance. An example of a hard RTS is a pacemaker. If the system takes longer than expected to initiate treatment, biological damage could result.

- In some types of (**firm**) RTS, timing requirements are a combination of both hard and soft ones. Typically the computation will have a shorter soft requirement and a longer hard requirement.
- example of a firm RTS is a patient ventilator that assists patients with breathing problems to breathe for them. The system must ventilate a patient so many times within a given time period. A few seconds delay in the initiation of the patient's breath is allowed, but no more than this.

Control Systems

Control Systems (for Physical World Tasks)

- The simplest type of control is activated only when defined thresholds are crossed e.g., a thermostat switches the heating on when the temperature falls below the lower threshold or switches the cooling on when it rises above the upper threshold. However, this frequently leads to oscillation of the temperature between the lower and upper thresholds.

➢ Programmable Controllers

- Programmable controllers have been developed to support much more configurable and flexible control, e.g., microcontrollers.
- The hardware architecture of microcontrollers is much simpler than that of the more general purpose processor mother boards in PCs.
- Microcontrollers do not need to have an address bus or a data bus, because they integrate all the RAM and non volatile memory on the same chip as the CPU.
- The CPU chip can be simpler, 4 bit processors as opposed to 64 bit processors. I/O control can be simpler as there may not be any video screen output or keyboard input.
- However, micro controllers can range in complexity, for example, they can include digital signal processors (DSPs). Originally, microcontrollers were only programmed in assembly language and then later in C code.
- Programs are often developed in an emulator on a (PC) development platform that is then downloaded to the target device for execution, maintenance, debugging and validation.

➢ Simple PID-Type Controllers

- A PID controller combines Proportional, Integral and Derivative type control.

- A proportional controller is just the error signal multiplied by a constant and fed out to a hardware drive. An integral controller deals with the past behaviour of the control.

- It is almost always used in conjunction with proportional control. It sums all the preceding inputs and is used to add longer term precision to the control loop by making use of the past history.

- A derivative controller is used to predict the plant behaviour, then this might be used to stabilise the control via a feedback loop. Such control algorithms, P, PI, PD or PID, are often simple enough to be hard coded into controllers along with some adjustment controls, e.g., the gain of the proportional element.

➢ More Complex Controllers

- The PID controllers described in the previous section can be used for coarse control of robots' arms, e.g., to support palletising, and to support coarse controlled locomotion.

- The control design algorithms are based on the assumption that some basic knowledge exists of the plant to be controlled was available and that some approximating models can be used to form a starting point for the design of the controllers and can then be tuned using feedback to cope with the uncertainties of the plant parameters.

However, the algorithms are not well suited to tasks that need finer control, dynamic control, dealing with uncertainties in the control and the need to support adaptive control.

There are several main sources of uncertainty in control that can occur:

• *environment dynamics* is a stochastic form of physical objects that naturally varies;

• *random action effects*: some actions can have different effects because of variations in physical characteristics of things when they are formed, complex effect propagation;

• *inaccurate models*: some characteristics of the physical environment, e.g., the weather, are inherently non deterministic;

• *approximate computation*: e.g., non linear behaviour is often approximated by a linear solution in a part of its range;

• *sensor limitations*: improper sensor placement, partial views of world, and poor signal reception,

- Two of the main techniques for controlling uncertain systems are **adaptive control and robust control**.

- Adaptive control uses online identification in which either the plant parameters of interest are identified using error predictions (indirect adaptive control), or the parameters are monitored and are adjusted by tracking errors (direct adaptive control) in an attempt to learn the uncertain parameters of the system.

- Adaptive control is a nonlinear feedback technique which performs identification and control simultaneously in the same feedback loop. This can be used to deal with systems with large uncertainties .

- The adaptive approach is applicable to a wide range of parameter variations, but is sensitive to unstructured uncertainties. A robust controller is designed to make the system insensitive to all uncertainties and the final controller has a fixed structure. A robust controller is suitable for dealing with small uncertainties

Unit 2

Smart mobile, cards and device network

Introduction

Smart user devices in this unit focus on tab or card and pad sized devices in which the locus of control for the user interaction resides in user customisable systems. Mobility is an important feature of these smart devices.

Overview

- With overview of the different notions of mobility, This section continues by discussing smart mobile services as an extension of distributed service models.
- operating system support for mobile computers and communication devices is discussed, including how to handle resource constrained ICT resources and power management.
- Then smart card type device are discussed, these are much more resource constrained and application specific than smart phone type devices.
- Finally, some types of device networks are covered.

Smart Mobile Devices, Users, Resources and Code

- Users are naturally mobile, e.g., users can move in between Internet nodes, to log on and to access Web based content and email, anywhere, anytime.

- Users can carry personalised mobile networked devices with them to access services filtered according to their personal preferences and to be aware of their location context and adapt to it.

- Other types of inanimate hosts such as transport vehicles can also act as mobile hosts. Each of the main components of a UbiComm system can be mobile virtual processing and services at the operating system and application level, code, access devices, hardware and data resources.

Mobile Service Design

- Mobility service design builds upon the basic design of smart devices and services but is a more specialised variant of it.

- In order to simplify access by applications and users, mobility, e.g., locating and addressing mobile users and routing data to mobile receivers, should be designed to be more transparent to applications and users.

- There are three kinds of transpar ency for mobile services:

(1) user virtual environments (UVE)

(2) mobile virtual terminals (MVT)

(3) virtual resource management (VRM).

- **UVE** provides users with a uniform view of their working environments independent of the location and terminal type, e.g., Virtual Home Environments or VHE .

- **MVT** preserves the terminal execution state for restoration at new locations, including active processes and subscribed services, e.g., mobile agents .

- **VRM** permits mobile users and terminals to maintain access to resources and services by automatically requalifying the bindings and moving specific resources or services to permit load balancing and replication.

- Mobility transparency is more challenging across heterogeneous networks.

- A seamless handover from one type of network to another with a limited delay could be performed by application processes but this is easier if this is OS supported.

- There must be a smooth transition between being used on the network and being a self sufficient device.

- mobile phone processes should support a volatile connectivity model.

- Mobile devices can create new local data that may be business sensitive or personal. A Denial of Service (DoS) can occur when a mobile device is stolen or left behind.

- There are two possible solutions to handle DoS

➢ First, back ups, occasional wireless synchronisation or wired synchroni sation of data can be used.

➢ Second, a remote access model can be used to support a virtual distributed UVE type desktop on the mobile device so that data that appears to be local is actually managed remotely

- Mobile service design consists of various parts:

➢ SMS and Mobile Web Services

➢ Java VM and J2ME

➢ .NET CF

SMS and Mobile Web Services

- One of the earliest and but still most widely used data application for many mobile phones users is the **Short Messaging Service (SMS)**.

- SMS is a messaging transport service that supports a reliable two way, connection less, messaging protocol.

- Its main limitations are it is text oriented, that text size is limited to 160 characters or less, message latency is one minute or more, there is poor security at the application level and it is difficult to link messages to interactions and transactions.

- Because of the limitations of SMS, a more flexible data protocol, the **Wireless Application Protocol (WAP)** was introduced.

- WAP supports connection oriented, interactive communication sessions and multimedia.

- There is no hard message size limit, it has security, and latency is less than SMS.

- WAP supports data formatting and navigation for small screens.

- WAP was designed for sending data over wireless with delays, slow links and low resource mobile terminals.

- Mobile Web Service design often uses a three tier **thin client, client proxy, server architecture.**

- The thin client architecture simplifies the features on mobile devices: they are just used for information access and can run a mini Web browser or micro browser .

- A client proxy is used in order to offload handling the heterogeneity of adapting content to heterogeneous terminals, micro browsers and content languages.

Java VM and J2ME

- J2ME, the Java 2 Mobile Environment, supports the development of fat client server system and standalone system designs in which code is developed.

- The mobile device can be networked to servers that are implemented using J2SE (Java Standard Edition) or J2EE (Java Enterprise Edition).

- In Java API , the same code can be written once and read and executed on many different devices.

- Rather than being structured as a single, monolithic platform with a large footprint for devices, J2ME is a multi-layered organisation of software bundles of three types:

➢ Configurations

➢ Profiles

➢ optional packages.

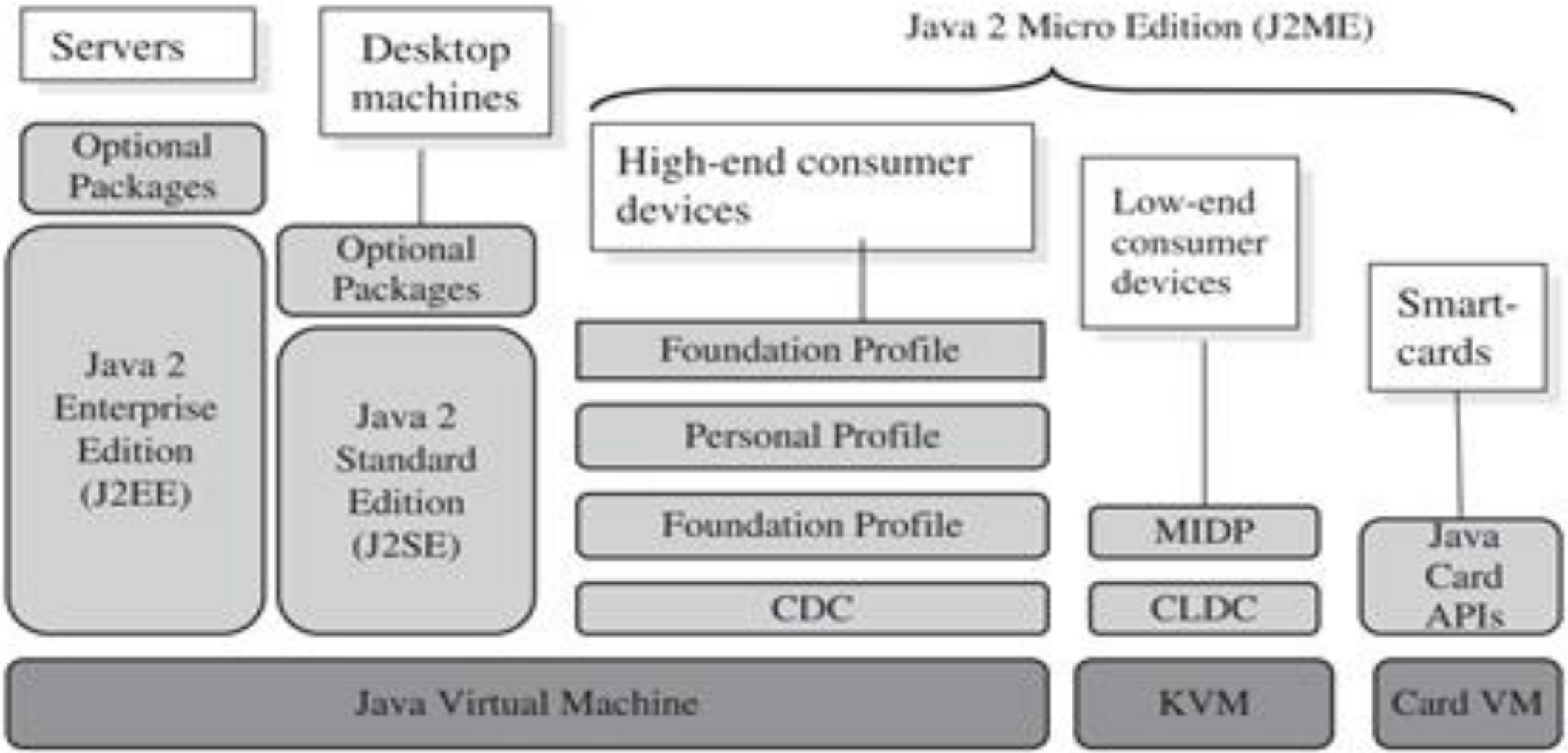


Figure 4.2 J2ME uses a VM to support a variety of devices

- A **configuration**, such as CDC or CLDC, provides fundamental services for a broad class of devices and virtual machines.

- A **profile**, such as the CLDC Mobile Information Device Profile MIDP, supports higher level services for more specific class of devices

- . An **optional package** adds specialised services that are useful on devices of many kinds, but not necessary on all of them,.

- J2ME platform delineates devices into two distinct categories or configurations

➢Connected Device Configuration (CDC)

➢Connected Limited Device Configuration (CLDC) devices

- **Connected Device Configuration (CDC)** devices, These are devices with more than 2MB of both RAM and ROM and support constantly connected networks that have a special purpose or are limited in function. They are not general purpose computing machines, e.g., set top boxes, Internet TVs, Internet enabled screen phones, high end communicators, and car entertainment/navigation systems. These devices generally have higher resource UI facilities and have more computing power.

- **Connected Limited Device Configuration (CLDC)** devices, These are devices with 160 512 KB memory and support personal, intermittently connected mobile information devices with a limited GUI, e.g., mobile phones, two way pagers, PDAs, and organisers. CLDC also supports card based ICT devices with limited ICT resources and without a GUI.

.NET CF

- .NET Framework is an integral Microsoft Windows component for building and running software applications.

- The .NET Framework consists of two main parts:

➢ The common language runtime (CLR)

➢ Unified set of class libraries such as ASP.NET for Web applications and services, Windows Forms for smart client applications, and ADO.NET for loosely coupled data access.

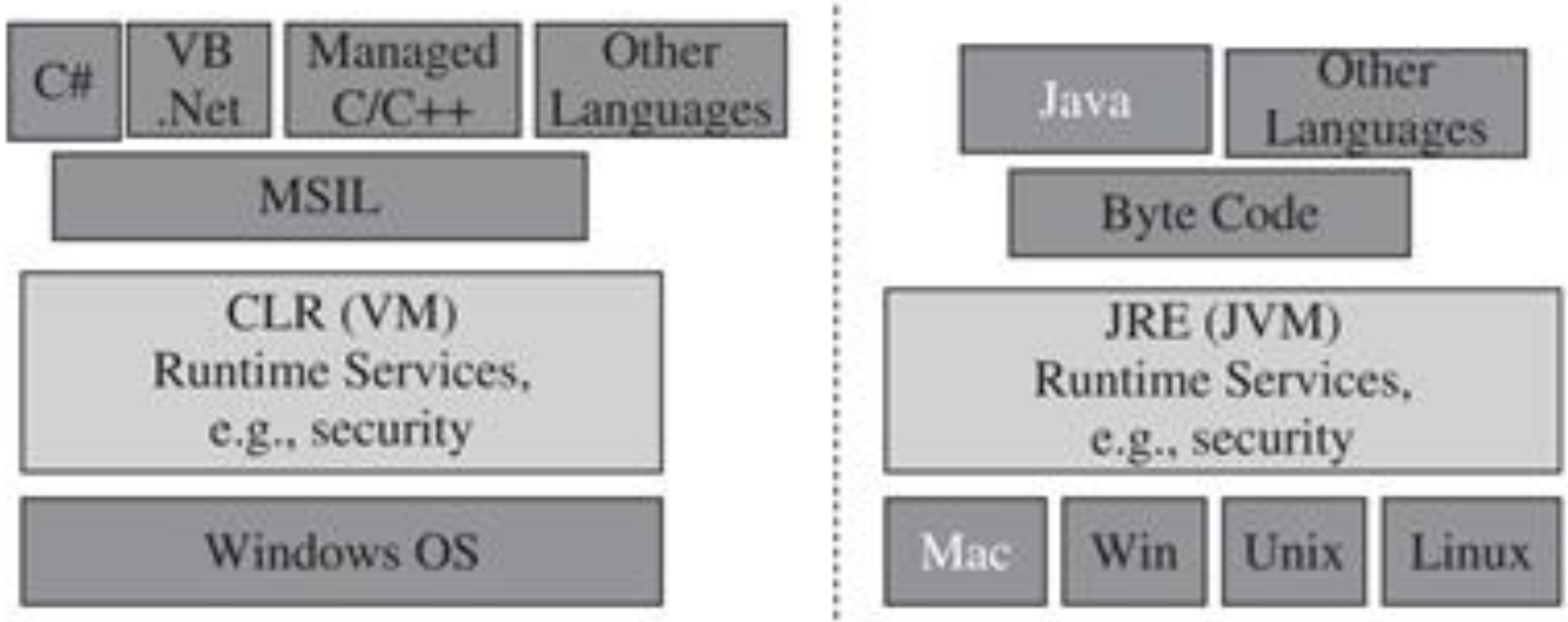


Figure 4.3 .NET VM versus the JRE VM

- The Microsoft .NET Compact Framework (CF) is a rich subset of .NET that is designed specifically for resource constrained devices, such as PDAs and smart mobile phones.

- Its class libraries are a subset of .NET. It also has a few additional libraries that are specific to mobile devices: IrDA support, SQL Server CE and device specific controls.

- .NET CF runs on a high performance JIT Compiler for mobile devices.

Mobile Code

- Code is usually designed to be downloaded from a remote service point. Installation requires configuring the code installation onto each platform.

- Once it is installed, it often remains at one point in the ICT infrastructure, although it can be maintained and updated.

- Unlike mobile computing, in which devices move or users move, and unlike static code where the code resides in one location, mobile code changes the network node where the program is executed.

- Code mobility is an important enabler for system extensibility, to support operation in open dynamic environments, in particular using resource constrained access devices.

- There are currently four main approaches to mobile code security :

➢ Sandboxes that limit the local services that code can access

➢ Code signing which ensures that code originates from a trusted source

➢ Firewalls that limit the machines that can access the Internet

➢ Proof carrying code (PCC) that carries explicit proof of its security.

Mobile Devices and Mobile Users

- Device mobility can be viewed from several dimensions

1. In terms of physical dimensions)

2. In terms of whether or not the device is mobile or some kind of host to which it is attached to is mobile

3. In terms of what kind of host, mobile devices can be bound to

4. In terms of how devices are attached to a host

5. In terms of when the mobility occurs. Each of these is discussed in turn.

- Mobility depends upon physical dimensions, the smaller a device is, the less energy is required to move it, increasing the degree of mobility.

- Devices may be intrinsically mobile using their own rechargeable or renewable power sources to drive them, e.g., a robot.

- Devices may be mobile because they are attached to some host which may be mobile. A mobile host may be animate or inanimate.

- Devices may be mobile because they are bound in some way to some embodied mobile host such as a living thing or some physical world artefact such as a vehicle or they can be attached to non living things which move

- There are three basic ways mobile devices can be physically bound to mobile hosts: **accompanied**, **surface mounted** or **embedded** into the fabric of a host, e.g., an embedded controller embedded in a host device.

- Accompanied refers to an object being loosely bound and accompanying a mobile host, e.g., a mobile phone can be carried in a bag or pocket but which can easily be misplaced.

- A device can be surface mounted onto a mobile host. When the mobile host is a human, we refer to surface mounting as wearing. Smart paint can be sprayed onto a surface .

- Embedded means a device can become permanently attached inside a mobile host, e.g., a braking control system embedded in a vehicle. When a device is attached in this way to a human or animal, we refer to this as an implant.

❑ Operating Systems for Mobile Computers and Communicator Devices

Desktop computers and high ICT resourced mobile devices such as smart phones (fat client devices) often require more complex multi process control and the use of a Multi Tasking Operating Systems (MTOS) to support the execution of multiple users and applications.

Microkernel Designs

- In a macrokernel or Monolithic Kernel Operating System all the system utilities such as hardware related drivers, memory management, process support, process scheduling, network protocol stack and file system are in one, single, large, kernel system.
- The main benefit of the monolithic kernel system is that it is more efficient for a single processor system because fewer context switches are needed.
- Context switches are only needed between processes and the kernel utilities and not generally between processes.
- The main drawback of a monolithic kernel system for low resource systems such as mobile systems is that use of the kernel is quite large and requires many system resources. In addition, the monolithic kernel is potentially more complex and has more points of failure and can require more updates, e.g., in order to support adding new hardware.

- In contrast, in a **microkernel**, only the fundamental parts of the operating system such as basic memory management, (limited) process management and inter process control are supported.

- The potential benefits of a small kernel are that it is more manageable in a low resource environment and more robust. It can still function even when system utilities, not in the kernel, fail.

- The drawbacks are that there is potentially more context switching between application, non kernel utility and the kernel utility process execution contexts, thus potentially lowering the performance when run on single processor systems.

Mobility Support

- Some original designs for OS for low resource device such as mobile devices were based on creating a cut down version of PC type MTOS, e.g., Windows Mobile, while other designs for an OS were specifically oriented to a lower resource mobile device and other specialist characteristics from the ground up.

- Mobile device design is still evolving. Some aspects of mobile OS support are even being proposed to produce low cost, low resource, low power, PC design.

- In addition, to the normal OS ICT support, mobile devices have several more specialised requirements including HCI for small mobile user interfaces , heterogeneous communication support, intermittent connections , data management , mobility support and power management.

Resource-Constrained Devices

- OSs for resource constrained devices, e.g., hand held mobile devices, must cope with a low primary memory, low secondary storage capacity, slow CPU, limited input/output and operating in a low power mode because of a finite energy supply.

- Strategies to cope with low memory include data compression, offloading data storage from a device (thin client) to remote networked servers, simply over writing older data and using larger capacity secondary memory.

- Strategies to adapt computation to a slow CPU include offloading computation from local to remote networked servers, the use of variable voltage CPUs and the use of energy based and predictive process OS scheduling .

- Strategies to decrease the energy required for communication involve using communication strategies where higher power servers and other devices initiate transmissions and the use of short range transmissions.

- Satyanarayanan (2001) proposes different strategies to adapt to resource constraints.

- **First**, the system can automatically guide applications to modify their behaviour so that they use less of a scarce resource. This change usually reduces the user perceived quality, or fidelity, of an application.

- **Second**, a client can ask the environment to guarantee a certain level of a resource. This is the approach typically used by reservation based quality of service. From the view point of the client, this effectively increases the supply of a scarce resource to meet the client's demand.

- **Third**, a client can suggest a corrective action to the human user of the application and it is up to the user to follow this through or not. UbiCom system adaptation in general is considered in more detail as part of context awareness

Power Management

- Power management is crucial for many embedded and mobile devices that are not attached to a permanent power supply.
- It increases their operational time in the field before their energy supply needs to be replenished and it reduces the time and effort needed to periodically replenish or replace nonrenewable energy sources.
- Power management is also a concern for devices permanently connected to a power source because of environmental and economic concerns.
- There are various strategies for device power reduction including:

➢ Manual overrides for users to specify when they no longer need to use devices

➢ The use of simple time outs to detect long idle states and to power down devices to a sleep mode or off mode

➢ The use of more finely grained techniques than on sleep off modes for power consumption and power conversion to store energy for later use that would otherwise be wasted.

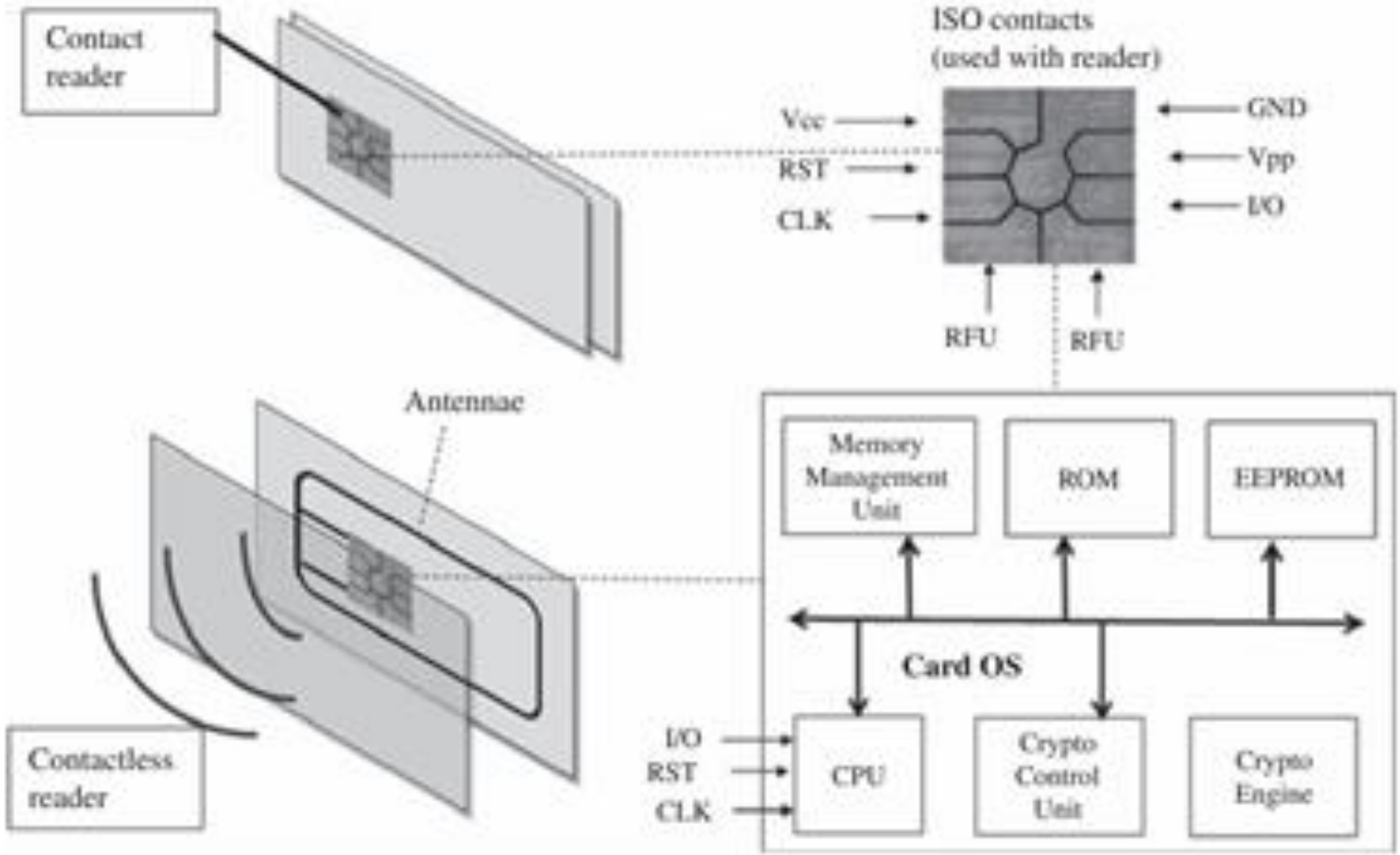
❑ Smart Card Devices

- A smart card is a plastic card embedded with digital memory and a microprocessor chip, as opposed to cards which store data on magnetic strips.

- It is reprogrammable, stores and processes data in the card and transacts data between card users and applications.

- Data can be stored and processed within the card's memory or microprocessor, which is accessed using a card reader.

- Smart cards are small and easy to carry around. They provide a secure data container, can be used for authentication purposes, e.g., as a hardware based digital signature, and can be used for metered services


Contact reader
ISO contacts (used with reader)
Vcc
RST
CLK
GND
Vpp
I/O
RFU
RFU
Antennae
Memory Management Unit
ROM
EEPROM
Card OS
Contactless reader
I/O
RST
CLK
CPU
Crypto Control Unit
Crypto Engine

- Many things have the potential to be replaced by a smart card, including driver's licence, insurance information, chip and pin credit or debit bank card, travel card and ticket, etc.

- Smart cards potentially represent a virulent form of Privacy Invasive Technology or PIT.

- Smart cards maybe either **contact** based or **contactless**.

- **Contact** cards have to be inserted in a reader and a physical electrical contact made. This may be chosen for security reasons, e.g., a chip (and enter) PIN (Personal Identification Number) application.

- **Contactless** cards also contain an antenna and an RF transceiver or transponder. The card can be waved in the immediate vicinity of a reader or base station in order to interact, e.g., a smart travel card designed to be contactless reduces the time to gain authorisation compared to the use of a contact card.

Smart Card OS

- The primary tasks of a smart card operating system on behalf of applications are transferring data to and from the smart card, controlling the execution of commands, managing files, managing and executing cryptographic algorithms to protect access to stored data, and managing and executing program code.

- The components of a typical card OS are combined into a single IC chip to reduce size.

- To standardise the communication protocol between the on card application, and off card programs accessed via the reader, the ISO/IEC 7816 standard has been defined.

Smart Card Development

- Java Card refers to the technology that allows small Java based applications (applets) to be run securely on smart cards and similar small memory footprint devices such as JavaRings.

- A Java Card is the tiniest version of Java targeted for embedded devices.

- It is widely used in SIM cards (used in GSM mobile phones) and ATM cards.

- Java Card applications use Java Card bytecode, a subset of standard Java bytecode, executed in a Java CardVM.

- Java CardVM runs in many smart cards, e.g., the GSM phone Subscriber Identification Module (SIM) card.

Four main steps comprise Java Card applet development using the JavaCard development kit:

- First, the applet functions are specified, e.g., a security function requires the user to enter a PIN; the card locks after three unsuccessful attempts to enter the PIN, etc.

- An ID (Applet IDs) is requested and assigned to both the applet and the package containing the applet class.

- The class structure of the applet programs is defined.

- The interface between the applet and the terminal application is defined.

❑ Device Networks

- The goal of a device network is to enable a wide variety of devices to interoperate.

- These activities include home automation such as light and climate control, person aware systems, home security, care in the community and pervasive AV content access.

- The enablers for this are home network and service infrastructures that are easy to install, to configure and to maintain; low cost (capital and operational cost) devices and infrastructures, and useful applications.

The home device infrastructure is highly fragmented and far from being a seamless infrastructure.

- First, many electronic devices are monolithic, may not be digital and are not network enabled.

- Second, heterogeneous control devices and networks pre dominate (InfraRed, X10, HAVi and HES , UPnP and Jini and OSGi).

- Third, inherent issues with wireless such as the ease of discovery and the finite power supply of many mobile devices are a major concern.

➢ HAVi, HES

- **Home Audio Video Interoperability** or **HAVi** is an industrial standard specification that provides standard APIs for typical audio and visual home appliances.

- The **Home Electronic System (HES)** is an international standard for home automation under development by experts.

HES defines the following system components:

- A Universal Interface is incorporated into an appliance for communicating over a variety of home automation networks.
- A Command Language for appliance to appliance communications is independent of which network carries the messages.
- A residential gateway, HomeGate, links home control networks with external service provider networks.
- Standard interfaces are defined to support interoperability among application domains, such as security, lighting, energy manage ment and for command, control, and communications in commercial and mixed use buildings such as apartment houses, retail shops and offices.

➢ Device Discovery

- If device networks are to support open, dynamic, heterogeneous device access and interoperability, automatic device discovery is necessary to simplify use.

- Device discovery standards and technologies include :

✓Sun’s Jini

✓UPnP forum headed by Microsoft

✓IETF’s SLP

✓DNS Service Discovery

✓Bluetooth’s SDP

Sun’s Jini

- Consists of three Java language protocols:

✓Discovery

✓Join

✓Lookup

- Discovery occurs when a service looks for a lookup service, either using a priori information or by using a multicast with which it can register.

- Join is used by service providers to register service capabilities.

- Lookup occurs when a client or user locates and invokes a service described by its interface.

UPnP forum headed by Microsoft

- Simple Service Discovery Protocol (SSDP) is a Universal Plug And Play (UPnP) protocol that uses HTTP notification announcements to request a service type URI and a Unique Service Name (USN).

- SSDP is supported in some firewall appliances so that applications can tunnel through HTTP to get external access and media exchange between host computers and media centres are facilitated using SSDP.

- The UPnP protocol is based upon HTTP/TCP or UDP/IP and XML.

- A control point in an UPnP network is similar to a client. It is capable of discovering and controlling other devices.

- UPnP uses plain XML.

- SSDP can work with or without its central directory service, called the Service Directory.

IETF's SLP

- IETF Service Location Protocol (SLP) provides a flexible and scalable framework to enable devices to access information about the existence, location and configuration of networked services.

- SLP provides a dynamic configuration mechanism for applications in local area networks and is designed to scale from small unmanaged networks to large enterprise networks.

- Applications are modelled as clients that need to find servers attached to any of the available networks within an enterprise.

- In cases where there are many different clients and services available, Directory Agents that offer a centralised repository for advertising services may be used. Some peripheral devices such as printers use SLP.

- In order to remain simple and flexible, SLP is designed to support service discovery but not to invoke services, unlike Jini and UPnP.

DNS Service Discovery

- DNS Service Discovery (DNS SD) is a way to use DNS to browse for services.
- It can be used with the multicast DNS (mDNS) protocol to discuss the network.
- In contrast to Microsoft’s SSDP, DNS SD uses DNS rather than HTTP.
- The DNS that DNS SD uses mDNS but unicast DNS can also be used.

Bluetooth's SDP

- Bluetooth Service Discovery Protocol (SDP) is specific to Bluetooth and supports search by service class and by service attributes, and service browsing when a Bluetooth client has no prior knowledge of the services available in its vicinity.

- If a Bluetooth device is set as discoverable, other devices can find it by broadcasting a query.

- Devices listed as 'not discoverable' do not respond to the query. If the search is successful, a device replies by sending the service record information,retrieved from its service discovery database, to the requesting device.

- The requester can then use the service record information to establish a connection with the other device.

➢ OSGi

- The Open Services Gateway Initiative (OSGi) defines and promotes open specifications, such as a core platform specification, for the delivery of managed services into networked environments such as homes and automobiles.

- The initial market for OSGi was home services gateways, e.g., in video broadcast set top boxes, broadband modems.

- These would then act as a gateway, between the end user (and owner) of the devices on a LAN, and the service providers, that could be accessible over the Internet who want to provide services for the devices such as home security, home health care monitoring and entertainment services

HUMAN–COMPUTER INTERACTION

Consist of various parts:

- The explicitness (eHCI) versus the implicitness (iHCI) of the interaction.
- The diversity of ICT device interfaces and interaction is considered starting with the interaction of four commonly used ICT devices: desktop and laptop PCs, mobile phones and games consoles.
- Then types of interaction and interfaces are considered further in a wider range of computer devices that support much more natural human computer interaction.
- Types of interaction and interfaces for human embedded devices are considered next.

Explicit HCI: Motivation and Characteristics

The basic concepts of HCI are:

- **Humans**: single or multiple users, with diverse physical and mental abilities, interacting coopera tively or competitively

- **Computers**: not just PCs but also a range of embedded computing devices and a range of device sizes such as dust, tabs, pads and boards.

- **Interaction**: may be directed via a command or by manipulating virtual objects (windows, desktop) but it can also involve more natural interaction such as speech interaction, gestures, etc

- HCI refers to the processes and the models for design and the operational interface for some products.

- For many users, the User Interface (UI) part of the system is the product.

- **Explicit HCI** puts the user at the centre of the interactive systems, so that the control of the system, responds to and is driven externally by the user, rather than the system being driven internally.

- Poorly designed UIs can lead to both higher training costs and higher usage costs and of course leads to lower product sales.

- The success of a product depends largely on both the user's experience with how usable a product is and how useful it is in terms of the value it brings to them.

Complexity of Ubiquitous Explicit HCI

Explicit HCI is complex for UbiCom scenarios even if it is well designed for individual devices

- because we need to use tasks as part of activities that require access to services across multiple devices
- because devices can be used by different types of people
- because users are engaged in multiple concurrent activities
- because users are engaged in activities which may occur across multiple physical environments
- because activities may be shared between participants
- because activities on occasion need to be suspended and resumed

Implicit HCI: Motivation and Characteristics

- It is defined as ‘an action, performed by the user that is not primarily aimed to interact with a computerised system but which such a system understands as input’.

- To support iHCI interaction is more about C2H (Computer to Human) Interaction.

- Implicit interaction is based on the assumption that the computer has a certain understanding of users’ behaviour in a given situation: it is a user aware type of context aware system .

- This knowledge of users’ behaviours in given situations is then considered an additional input to the computer while doing a task .

- Implicit interaction can allow some of the interaction to be hidden from users and hence for the device to become invisible.

- There are some obvious challenges in supporting implicit interaction.

- It can sometimes be complex to accurately and reliably determine the user context because of: the non determinism of the subject, the user has not clearly decided what to do, or because of non determinism in the subject's environment.

- Implicit interaction, in contrast to explicit interaction, reduces or even removes explicit user control.

User Interfaces and Interaction for Four Widely Used Devices

- Four of the most commonly used networked ICT devices are the personal computer in its various forms such as

✓desktop and laptop,

✓hand held mobile devices used for communication,

✓games consoles and remote controlled AV displays,

✓players and recorders.

- Each of these can be designed to perform a common set of functions such as AV player, record, output, data and AV communication and to act as a hub or portal for multiple service access and interoperability.

➢ Diversity of ICT Device Interaction

- The term computer in the HCI has a much more diverse meaning within the field of UbiCom.

- It refers to any device with a programmable IC chip inside, including a range of multi task operating system (MTOS) devices.

- Common multi task devices include desktop and laptops PCs, mobile phones, games consoles, AV recorders and players such as televisions, radios and cameras.

- Even more numerous than MTOS computers are computers embedded in devices that perform specialised tasks such as various household appliances, vehicle control systems, travel ticket machines, cash dispensers, building controls, etc

There are several dimensions devices could be characterised according to:

- **size**: hand sized, centimetre sized, decimetre sized versus micro sized versus body sized or larger.
- **Haptic input**: two handed versus one handed versus hands free operation
- **Interaction modalities**: single versus multiple.
- **Single user versus shared interaction**: in personal space, friends' space or public space.
- **Posture for human operator**: lying, sitting, standing, walking, running, etc.
- **Distance of output display to input control**: centimetres to metres.
- **Position during operation**: fixed versus mobile.
- **Connectivity** : stand alone versus networked, wired versus wireless.
- **Tasking** : single task devices versus multi task devices.
- **Multimedia content access**: voice and text communication oriented, alpha numeric data or text oriented, AV content access.
- **Integrated** : embedded integrated devices versus dynamically interoperable devices.

➢ Personal Computer Interface

- **MEMEX system** : ‘It consists of a desk, and while it can presumably be operated from a distance, it is primarily the piece of furniture at which he works. On the top are slanting translucent screens, on which material can be projected for convenient reading. There is a keyboard, and sets of buttons and levers. Otherwise it looks like an ordinary desk.’

- **WIMPS interface** : supports direct manipulation of visual objects prevailed. Users can interact with computers by typing text commands but for many tasks direct interaction can be used by activating and moving active window sub areas called icons and menus, using mouse clicks and mouse drag and drop respectively.

- **Dialogues** are mechanisms in which users are informed about pertinent information that they must acknowledge receipt of or they ask for input to constrain a query. Typically, this type of interface is displayed in the form of a pop up window called a dialog box.

➢ Mobile Hand-Held Device Interfaces

- PC style WIMPS interaction will not work as effectively on mobile devices because the display area is smaller.
- It is impractical to have several windows open at a time. It can be difficult to locate windows and icons if they are deeply stacked one on another and to resize them.
- Screen navigation using fingers on a touch pad or an external device may be too big and unwieldy for small devices.
- In addition, the keyboard is smaller for user input and there is a greater variety of input devices.
- Instead of using the inbuilt device interface, the device can be attached to different kinds of external input interface which are available in the environment.
- It could become common to have displays, keyboards and Internet work connections at fixed hotspots and to allow users to plug in their own mobile access device.

Handling Limited Key Input: Multi-Tap, T9, Fastap, Soft keys and Soft Keyboard

- **Multi-Tap**

A typical numeric keypad used in current standard mobile devices has 12 keys. Eight of the numeric keys have three alphabetic characters associated with them that can be triggered by multi tapping keys, e.g., for the 2 key, 1 tap gives ‘a’, 2 taps gives ‘b’, etc.

- **T9**

Multi tapping can be enhanced with the introduction of a dictionary based predictive text method known as T9 and can produce a text entry speed up to 40 words per minute. T9 is an example of implicit interaction complementing explicit interaction the user types the start of words, the system tries to automatically complete the rest of the words based upon a simple text context

- **Fastap keypad**

Involves two keypads, one with smaller keys raised at the corners above the other keypad keys. The upper one is used for alphabetic input, the lower one for number input, both laid out over the size of a business card. If several keys are hit at once, a technique called passive chording allows the system to work out what the user intended to enter.

- **Soft keys**

Enable the meaning of the two left and right keys at the top of keypad to be determined by information on the screen; this allows the same keys to be reused to support application and task specific choices. Instead of having two soft keys, a whole mini keyboard, a soft keyboard, could also be displayed if there is sufficient screen space.

- **Internal pointer devices**

Such as tracker pad, roller pads, mini joysticks or even keyboard arrow keys can be used to move the pointer on the screen. Screens may also be designed to be touch screens whose areas can be activated using some physical stick like pointer, pen or a finger .

- **Auditory interfaces**

With speech recognition enable the use of voice activation to accept incoming calls and to initiate outgoing calls. However, there are additional challenges for mobile users when: operating in variable physical environments such as a noisy background; users have limited attention; users are less able to attend to the phone sometimes when they are engaged in other activities.

Handling Limited Output

- There are multiple techniques for overcoming limited output. This was an issue for many earlier desktop systems and not just an issue for current mobile devices.

- Mobile devices need to be smarter at filtering information to minimise the information available for display.

- Mobile displays tend not to use a Single Window WIMPS system because of the lack of control in using a pointer to select parts of a screen on a small display.

- The single window or screen can be switched to other content using a common navigation bar on each page, or hypertext links can be used to support between page navigation while scrolling can be used to support within page navigation.

There are several well known approaches to deal with size differences between the size of visual content and the size of the display.

- If content is too large, it can simply be cropped or the content resolution can be reduced or a zooming interface can be used. Zooming (in and out) coupled with the use of scrolling (up and down) and panning (side to side) control enable users to view content in an interactive way.

- Another development here is a peephole display which uses sensors to act as a tangible UI. Sensors determine the position of the device in relation to the user. As the device moves towards or away from the user, its position changes and its display can be updated to display different information.

- Two further approaches to enable small form sizes to display large visual content are to use MEM type micro projectors or to use organic displays which can be folded during transit and unfolded for operation.

➢ Games Console Interfaces and Interaction

- Games consoles are an important driver and can contribute to UbiCom in a number of ways.

- They can introduce innovative ways of interacting with physical environments into the main stream and they can help to advance the use of augmented and the more general mediated reality environments.

- At the start of computer games development, computers had no monitors but used paper, punched cards and paper tape for input and output.

- Early video games were coin operated and contained simpler electronic logic such as counters as at that time the first microprocessors had not yet been invented.

- **Atari** was the largest consumer of N channel LSI (Large Scale Integration) electronics in the 1970s and 1980s, driving the demand for the ROM and RAM memory chips that then were produced in high volume at a lower cost per chip, enabling personal computers with memory to be priced lower.

- **Forster** (2005) has identified seven different generations of games consoles based upon the technologies they use.

- Current, seventh generation, game consoles include the Nintendo Wii which has deviated from the standard D pad interface,4 by using a one handed wireless wand type games console interface instead.

- It contains micro sensors in the form of accelerometers located inside the controller and an infrared detector to sense its position in 3D space.

- The main focus of many types of electronic games is the interface between the person, the game and an elaborate scoring system.

- This scoring system is often tuned so that as the game progresses, it is harder and harder to score points but the points amass larger values.

- The introduction of a more natural interface, such as the Wii wand, can make it easier for broader audiences to interact with the system and become immersed in the virtual game environment.

- One type of device for which a profusion of remote controls exists is the AV player recorder whether in the form of mobile camera, fixed set top box player, etc.

- There are several design and wider engineering issues here. There is a profusion of remote control devices which have overlapping features and in some cases several devices need to be orchestrated with respect to a common feature, e.g., setting the volume for a home entertainment centre.

- A universal localised remote control may be useful. Devices may have limited network connectivity and may not always be connected to an IP network.

- Device control could be usefully operated remotely, e.g., because someone has forgotten to set a timer recording but this may cause conflicts if the device is being used by another local user who wants to perform a different AV function.

- If devices were universally connected to a common network, their status and utility could be more easily remotely accessed and configured.

➢ Localised Remote Control: Video Device

- Many devices that are body sized or larger such as earlier mainframe and mini computers in the 1970s and 1980s, large TVs and AV projectors and cars .

- These often require greater remote and automated control to reduce the degree of manual interaction that would otherwise need to be situated at the device interface.

- Typically these devices are numeric keypads rather than keyboards. If users need to enter text, e.g., for a caption to an AV recording, then letters are selected by using cursor keys to navigate around a soft (screen) keyboard.

- Some large devices have no remote control partly because the operation of the device requires local manual input which cannot yet be robustly automated.

Ubiquitous Computing

Chapter 1

Ubiquitous Computing: Basics and Vision

Overview

- Living in a Digital World
- Modelling the Key Ubiquitous Computing Properties
- Ubiquitous System Environment Interaction
- Architectural Design for UbiCom Systems: Smart DEI Model

Ubiquitous Computing (UbiCom)

- A vision for computing to:
  - Enable computer-based services to be made available everywhere (Ubiquitous)
  - Support intuitive human usage
  - But yet, appear to be invisible to the user.
  - Also referred to as pervasive computing etc

Ubiquitous Computing

Definition

- The term ‘ubiquitous’, meaning appearing or existing everywhere, combined with computing to form the term Ubiquitous Computing (UbiCom).

- It is used to describe ICT (Information and Communication Technology) systems that enable information and tasks to be made available everywhere, and to support intuitive human usage, appearing invisible to the user.

Trend: smaller, higher resource devices

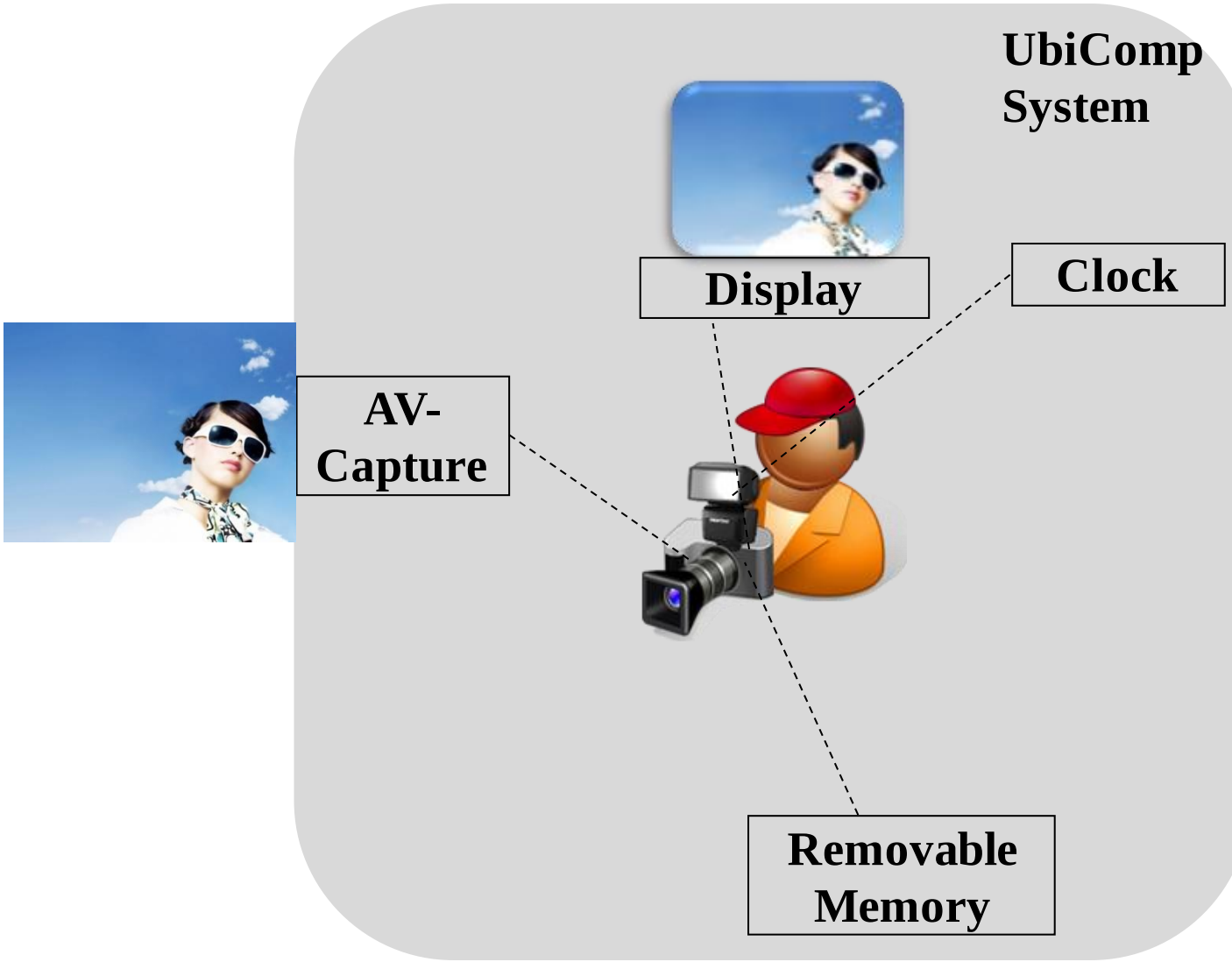


Scenarios

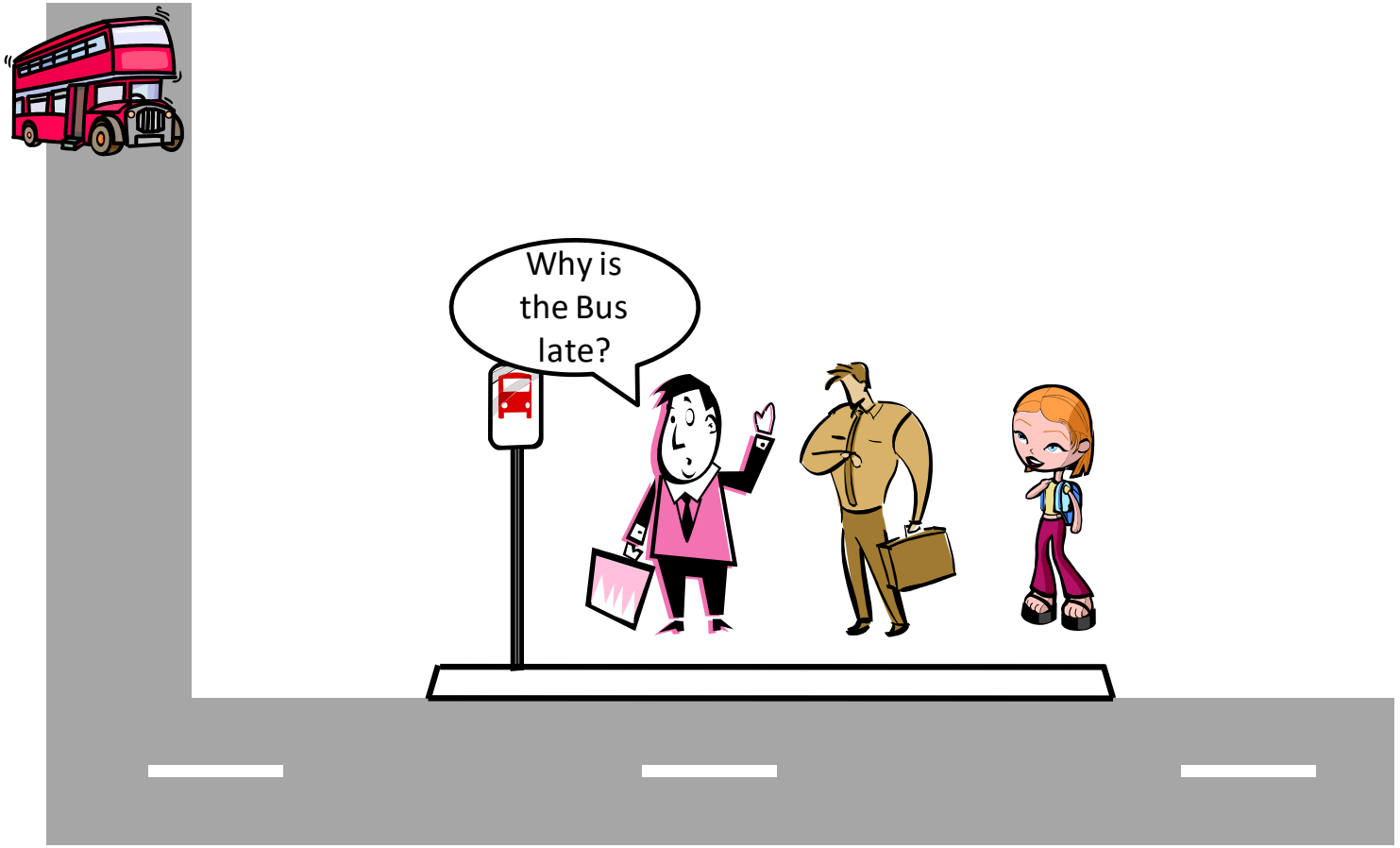
4 scenarios illustrate a range of benefits and challenges of ubiquitous computing:

- Personal memories
- 21st Century Scheduled Transport Service
- Foodstuff management
- Utility regulation

Personal Memories Scenario



21st Century Scheduled Transport Service Scenario



Foodstuff Management Scenario

Select & Buy food at physical or virtual market

Transport food to home store

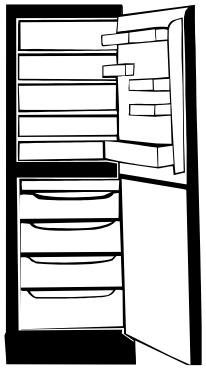
Put in home store

Select food from store, get & transform food into a meal

Consume food

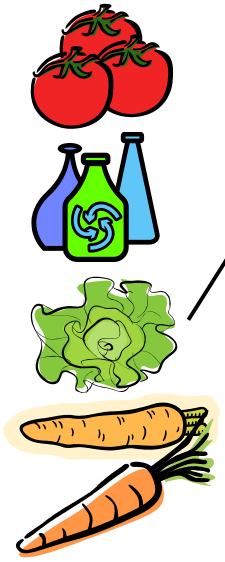






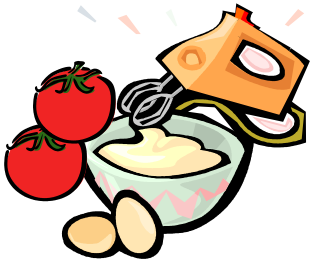






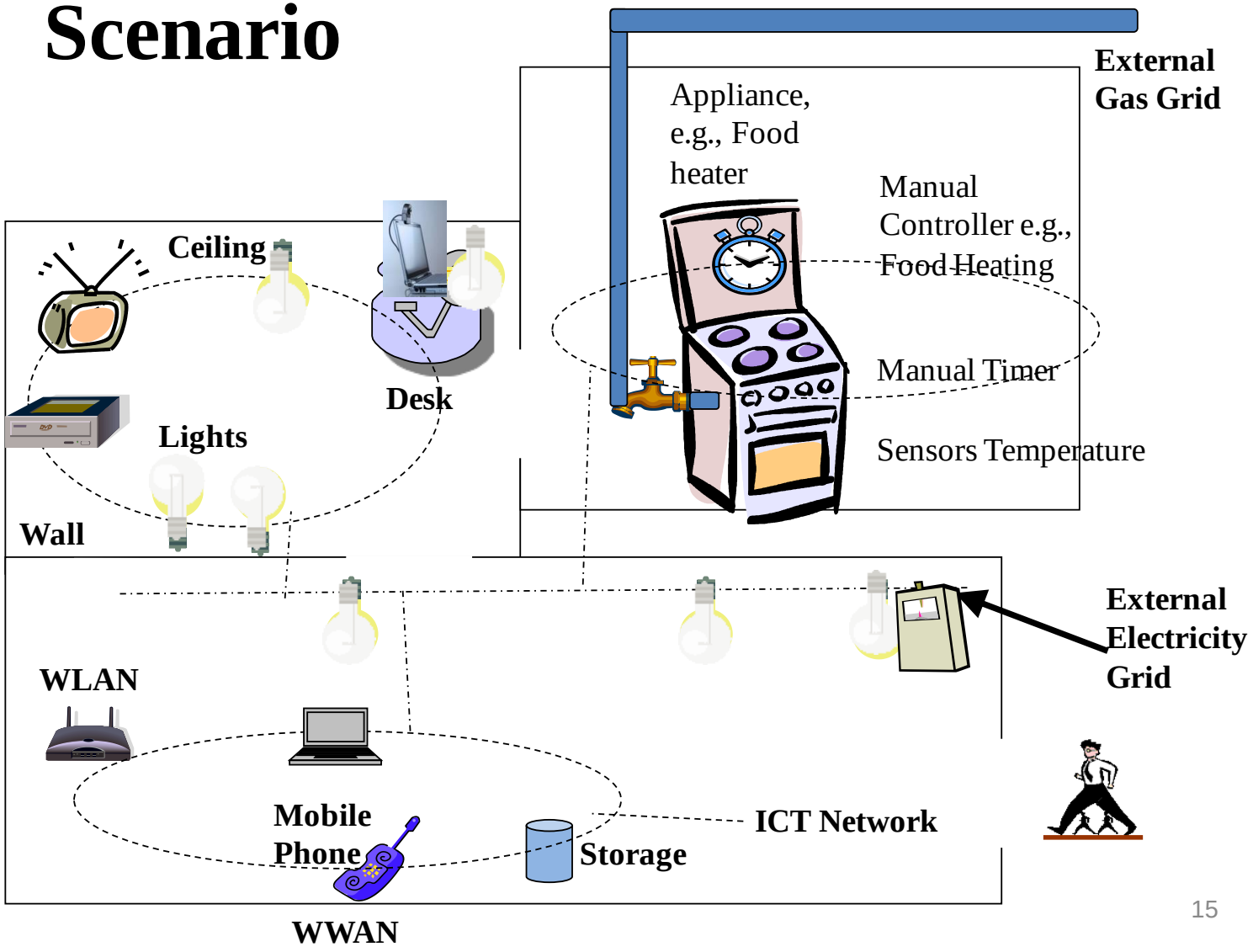








Utility Regulation Scenario



UbiCom System Design

For these scenarios

- Which system designs should be used for:
  - comms., data storage, processing, sensing, control etc

- How to model system - physical world interaction?

- How to model human computer system interaction?

- These are covered later in this course.

Framework for UbiCom : Smart DEI

- Design architecture to apply UbiCOm system :

  Smart devices, smart environment and smart interaction.

- Internal model of UbiCom system proterties based upon five fundamental properties:

  distributed, iHCL, context awareness, autonomy, AI

- A model of UbiCom system's interaction with its external environment.

Overview

- Living in a Digital World
- **Modelling the Key Ubiquitous Computing Properties**
- Ubiquitous System Environment Interaction
- Architectural Design for UbiCom Systems: Smart DEI Model

UbiCom: Different Combinations of Core Properties versus a Single Definition

- No single, absolute definition for ubiquitous computing.
- Instead propose many different kinds of UbiCom based upon combining different sets of core properties
- What core system properties would you propose to define ubiquitous computing?

.

UbiCom: Internal System Properties

3 main properties for UbiCom Systems were

1. Computers need to be networked, ***distributed*** and transparently accessible

2. Computer ***Interaction*** with Humans needs to be more ***hidden***

3. Computers need to be *aware* of ***environment context***

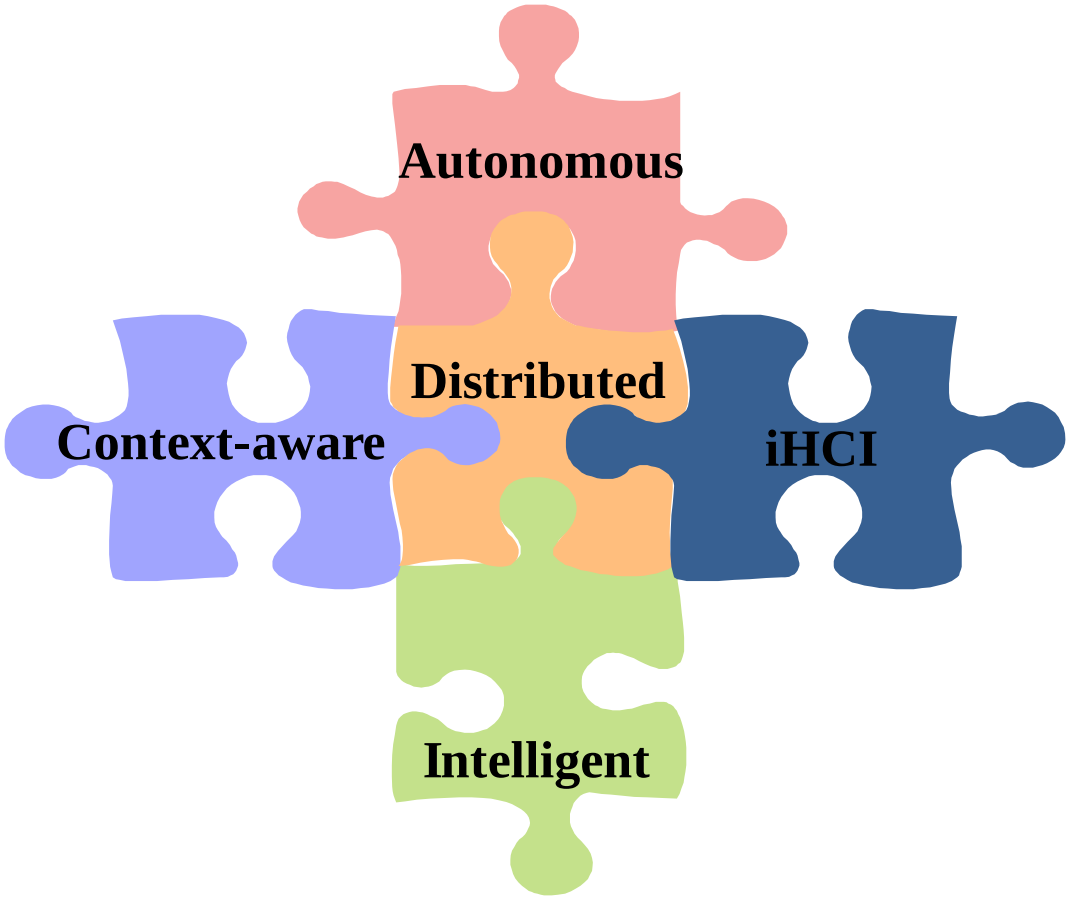
Devices: Extended set of Internal System Properties

To which two additional properties are added:

4. Computers can operate ***autonomously***, without human intervention, be self-governed

5. Computers can handle a multiplicity of dynamic actions and interactions, governed by intelligent decision-making and intelligent organisational interaction. This entails some form of ***artificial intelligence*.**

Five main properties for UbiCom



UbiCom System Properties: Distributed

- Networked ICT Devices
- Transparency
- Openness

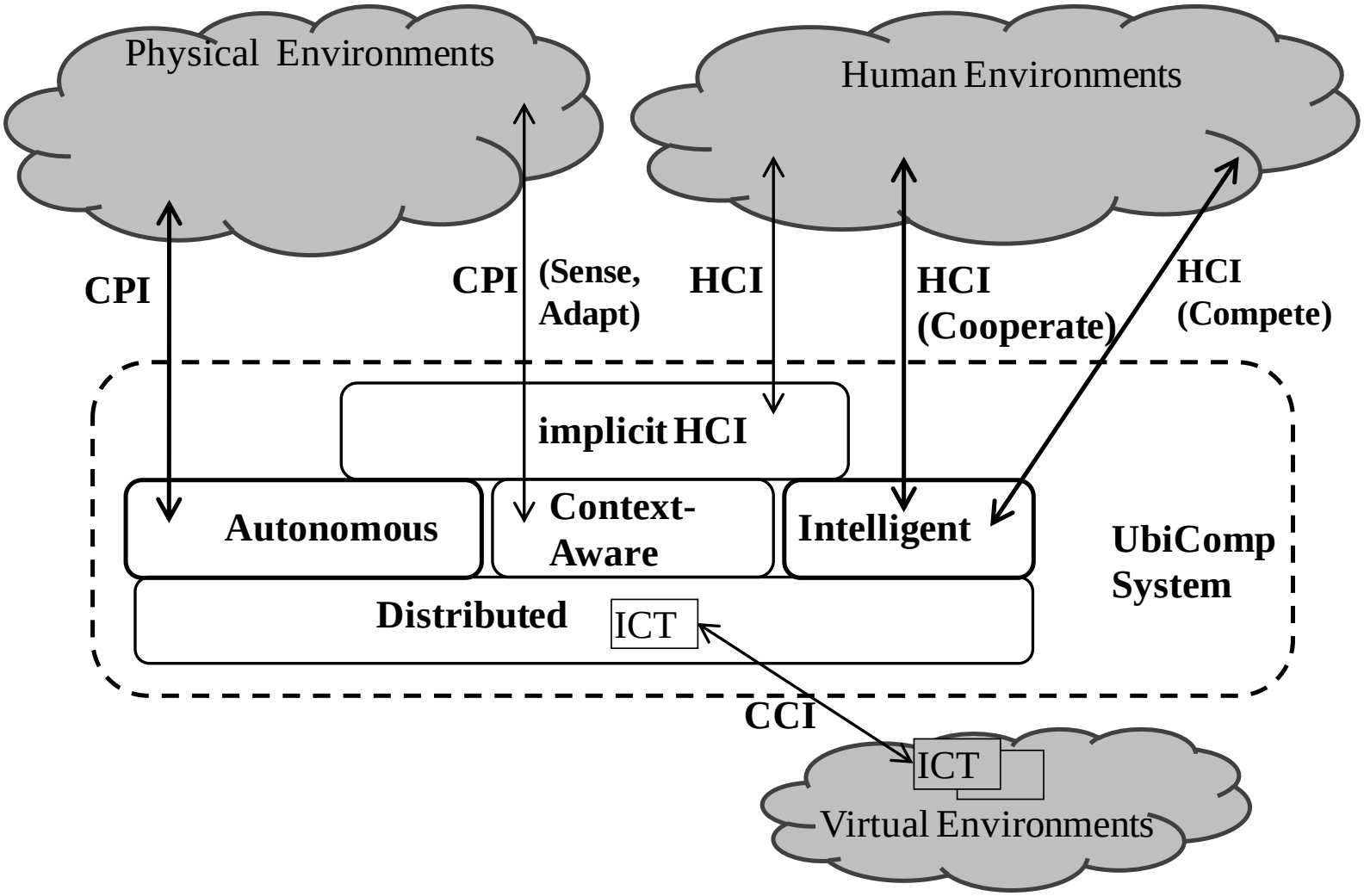
Distributed System: sub-properties

- Often designed middleware, set of generic services
- Universal, Seamless, Heterogeneous
- Networked
- Synchronised, Coordinated
- Open
- Transparent, Virtual
- Mobile, Nomadic

Internal System Properties: iHCI

- Concept of *calm* / *disappearing computer* has several dimensions
- Implicit (iHCI) versus Explicit HCI
- Embodied Virtuality as opposite of VR (people in virtual world)

Devices: Extended set of Internal System Properties



iHCI: Sub-Properties

Implicit Human Device Interaction (iHCI)

- Non-intrusive, Hidden, Invisible, Calm, computing
- Tangible, Natural
- Anticipatory, Speculative, Proactive
- Affective, Emotive
- User-aware
- PostHuman
- Sense of Presence
- Immersed, Virtual , Mediated reality

Internal System Properties: context-aware

3 Main Types of Context

- Physical Environment Context
- Human Context (or User context or person context)
- ICT Context or Virtual Environment Context

Context-aware: sub-properties

- Also referred to as Sentient, Unique, Localised, Situated
- Presupposes sensing of environment to be aware of its context
- Adaptive: active versus passive context-aware

Types of environment aware

- Person-aware, User-aware, Personalised, Tailored,
- Environment-aware, Context-aware, Physical context-aware
- ICT infrastructure aware

Internal System Properties: Autonomy

- Autonomy refers to the property of a system that enables a system to control its own actions independently.

- Autonomous systems are defined as systems that are self governing and are capable of their own independent decisions and actions.

- There are several different types of autonomous system. Ex : On the Internet, an autonomous system is a system which is governed by a router policy for one or more networks, controlled by a common network administrator on behalf of a single administrative entity.

Internal System Properties: Autonomy

- Reducing Human Interaction
- Easing System Maintenance Versus Self-Maintaining Systems

Autonomy: Sub-Properties

- Automatic
- Embedded, Encapsulated
- Embodied
- Resource-constrained
- Untethered, Amorphous
- Autonomic, Self-managing, self-star properties
- Emergent, self-organising

Internal System Properties: Intelligence

It is possible for UbiCom systems to be context aware, to be autonomous and for systems to adapt their behaviour in dynamic environments in significant ways, without using any artificial intelli gence in the system.

Intelligence can enable systems to act more proactively and dynamically in order to support the following behaviours:

- Modelling of its physical environment
- Modelling its human environment
- Handling incompleteness
- Handling non deterministic behaviour

Individual Intelligence: Sub-Properties

- Referred to as Intelligent Systems, AI, agent-based system etc.

Sub-Properties (sub-types of individual intelligence)

- Reactive, Reflex
- Model-based,
- Rule/Policy-based
- Logic/Reasoning
- Goal-oriented, Planned, Proactive
- Utility-based, Game theoretic
- Learning, Adaptive

Multiple Intelligence: Sub-Properties

- Referred to as Distributed AI, Multi-Agent Systems, Collective or Social Intelligence

Sub-Properties

- Cooperative , Collaborative, Benevolent
- Competitive, self-interested, antagonistic, adversarial
- Orchestrated, Choreographed, Mediated
- Task-sharing
- Communal, shared meaning
- Shared knowledge
- Speech-act based , Intentional, Mentalistic

Overview

- Living in a Digital World
- Modelling the Key Ubiquitous Computing Properties
- **Ubiquitous System Environment Interaction √**
- Architectural Design for UbiCom Systems: Smart DEI Model
- Course Outline

Ubiquitous System Environment Interaction

Three types of system environment

- UbiCom systems which form the ICT infrastructure, supporting services and act as middleware for that particular ICT system applications (virtual worlds)
- Human individuals and human organisations
- Physical world

Together, the virtual (computer) environment humans and the physical world can be considered as forming an external environment for UbiCom systems.

UbiCom system is organised as a layered information system stack

- Bottom layer of information resources
- Middle layer of processing
- Top layer of user information abstractions to view and interact with the information

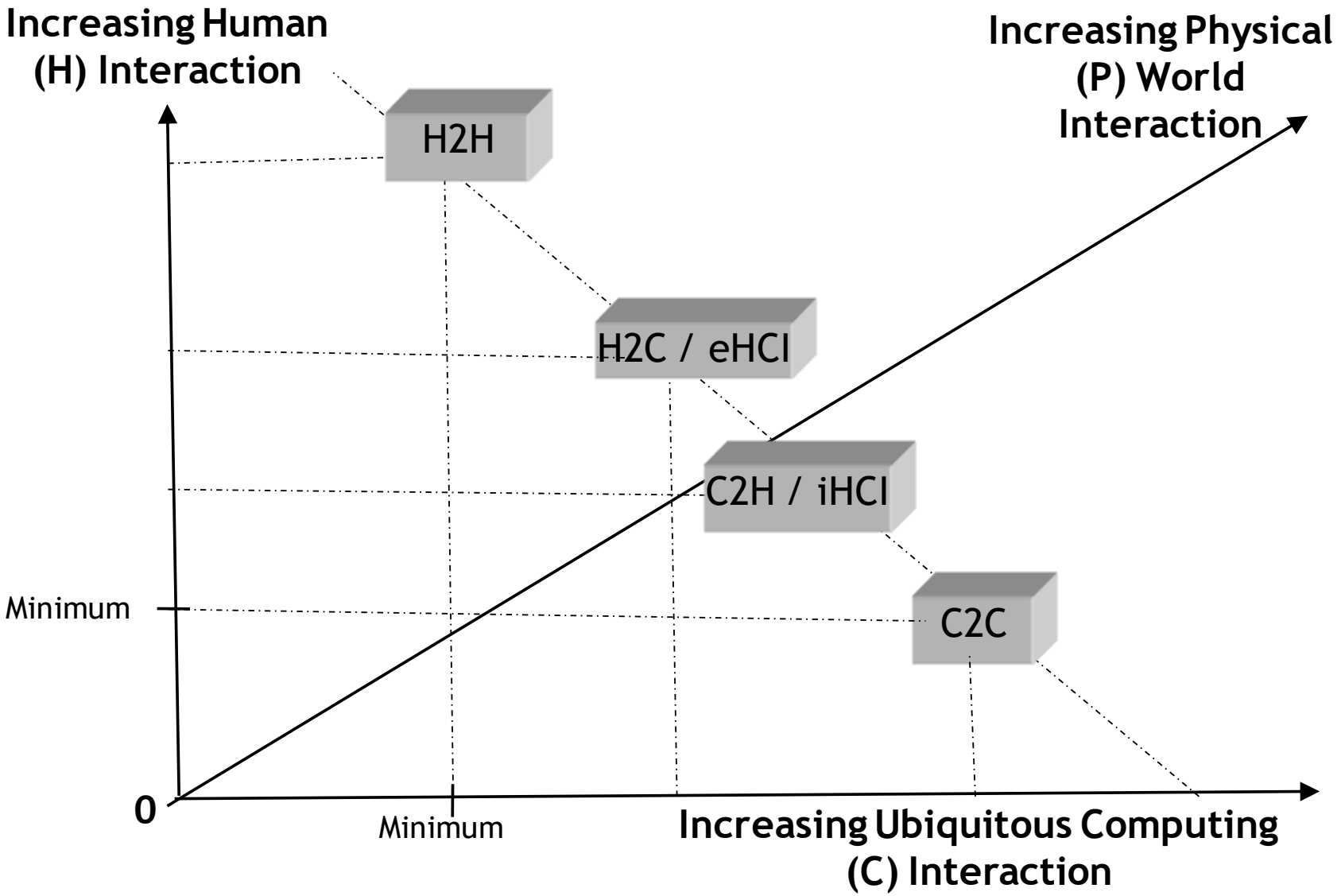
The physical environment can be represented using multiple models.

- local physical control model, e.g., lighting controls can sense the existing natural lighting and switch on artificial lighting when the natural light is below a certain threshold.
- A second type of physical world model is an ecology system.

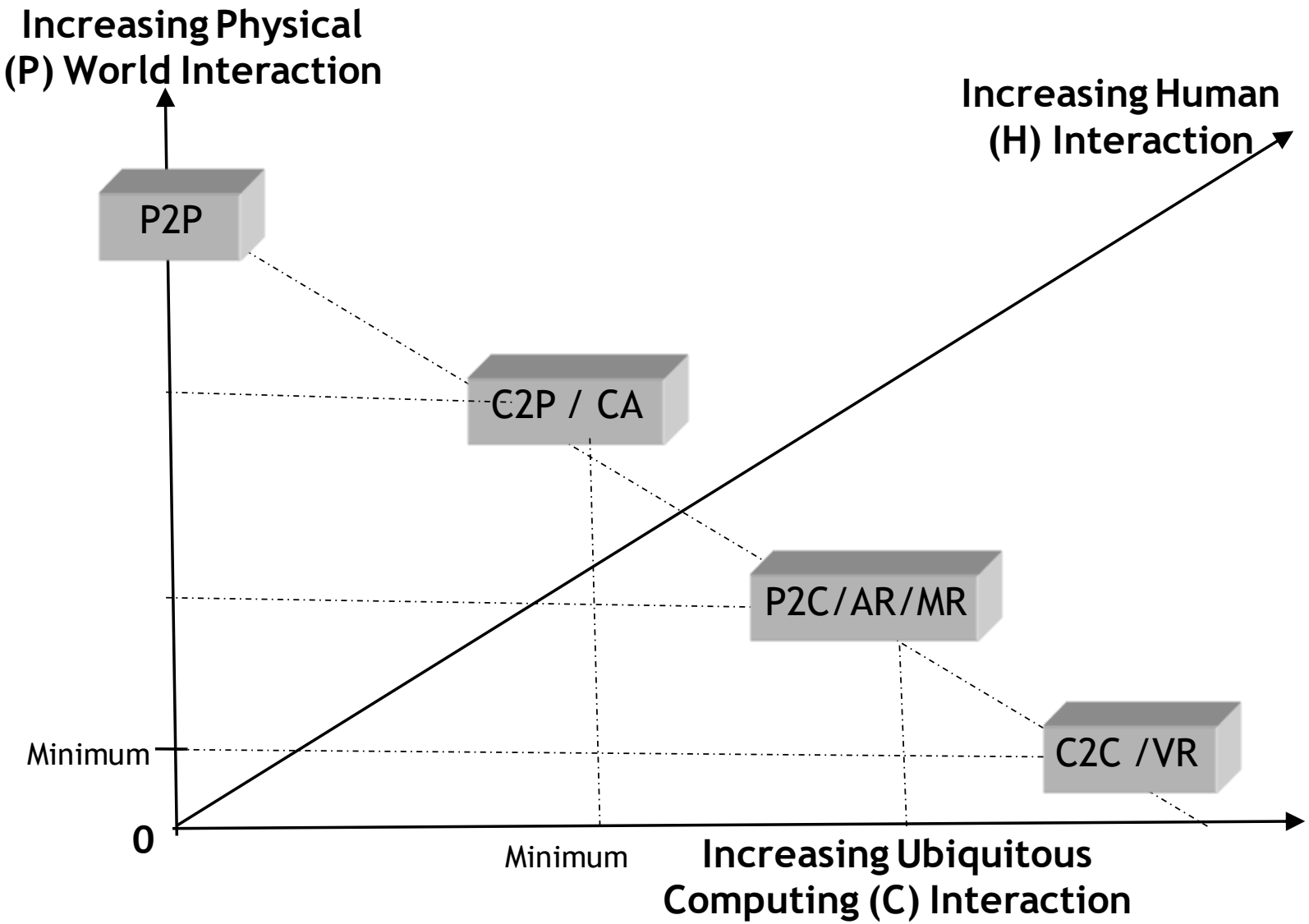
The system environment interaction occurs between

- Human and ICT system(HCI)
- ICT system and physical world (CPI)
- Between ICT system(C2C or CCI)

Different Degrees of HCI



Different Degrees of CPI

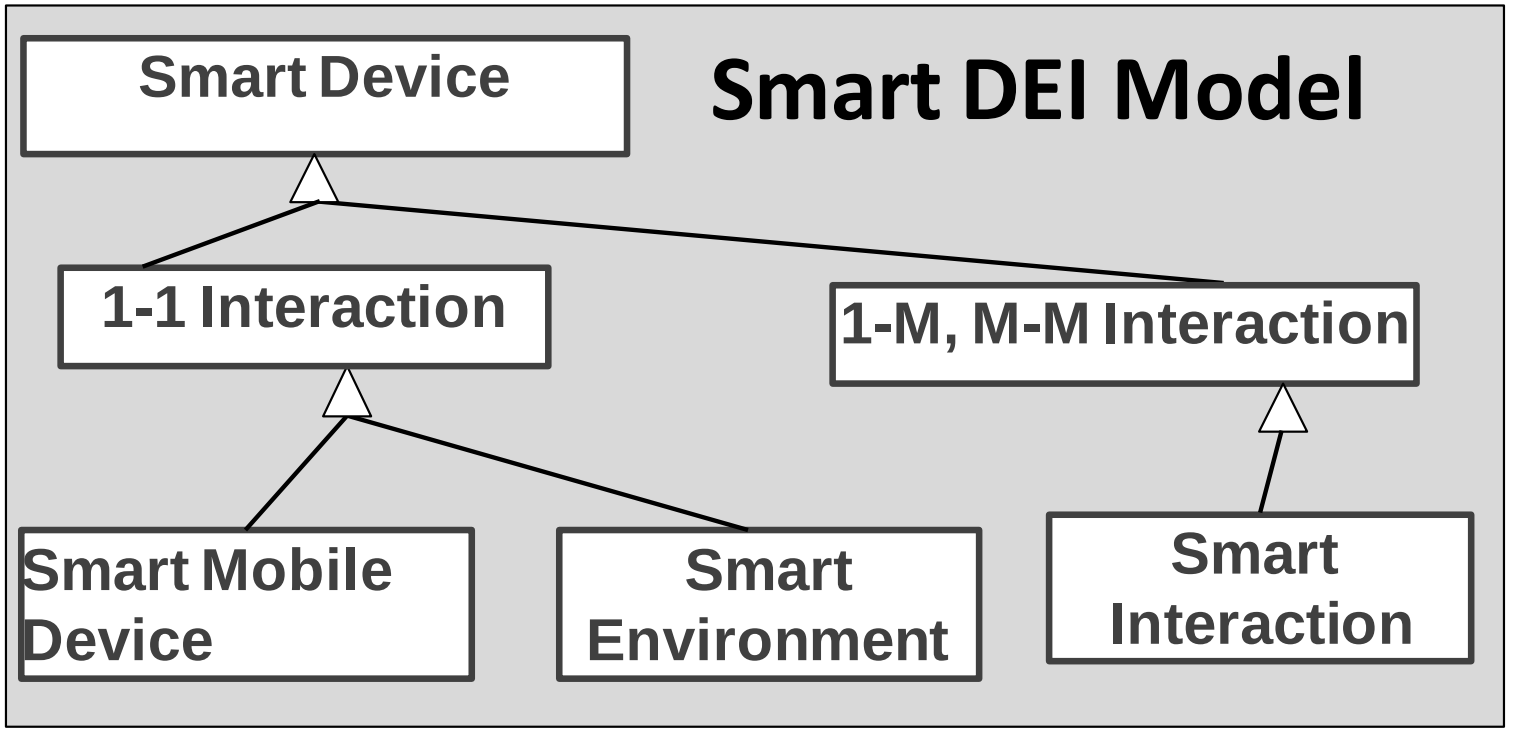


Overview

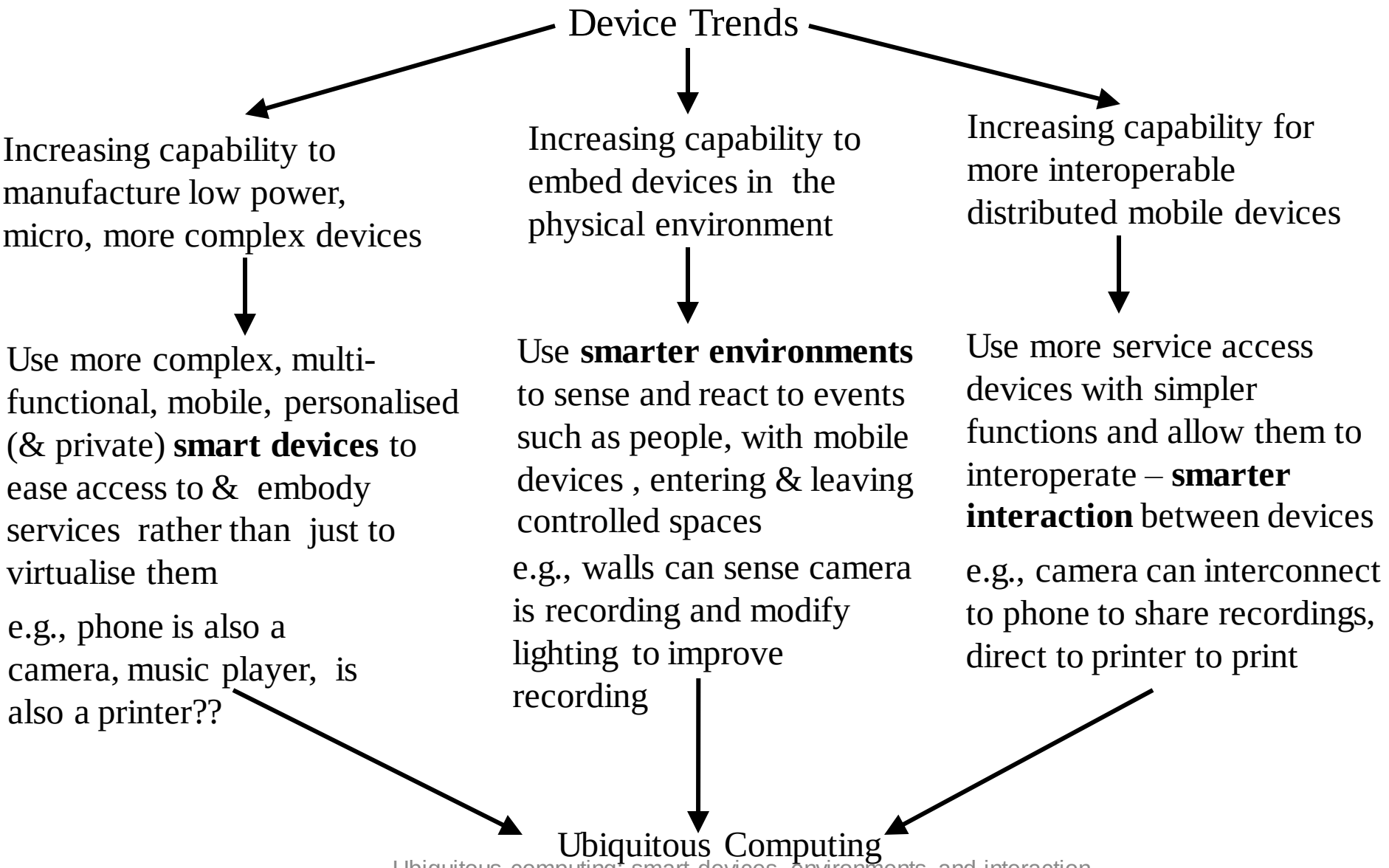
- Living in a Digital World
- Modelling the Key Ubiquitous Computing Properties
- Ubiquitous System Environment Interaction
- **Architectural Design for UbiCom Systems: √**
- Course Outline

UbiCom System Model: Smart DEI

- No single type of UbiCom system
- Different UbiCom systems support:
- 3 basic architectural design patterns for UbiCom:
  - smart **D**evices, smart **E**nvironments, smart **I**nteraction
- 'Smart' means systems are:
  - active, digital, networked, autonomous, reconfigurable, local control of its own resources, e.g., energy, data storage etc.


Smart Device
Smart DEI Model
1-1 Interaction
1-M, M-M Interaction
Smart Mobile Device
Smart Environment
Smart Interaction

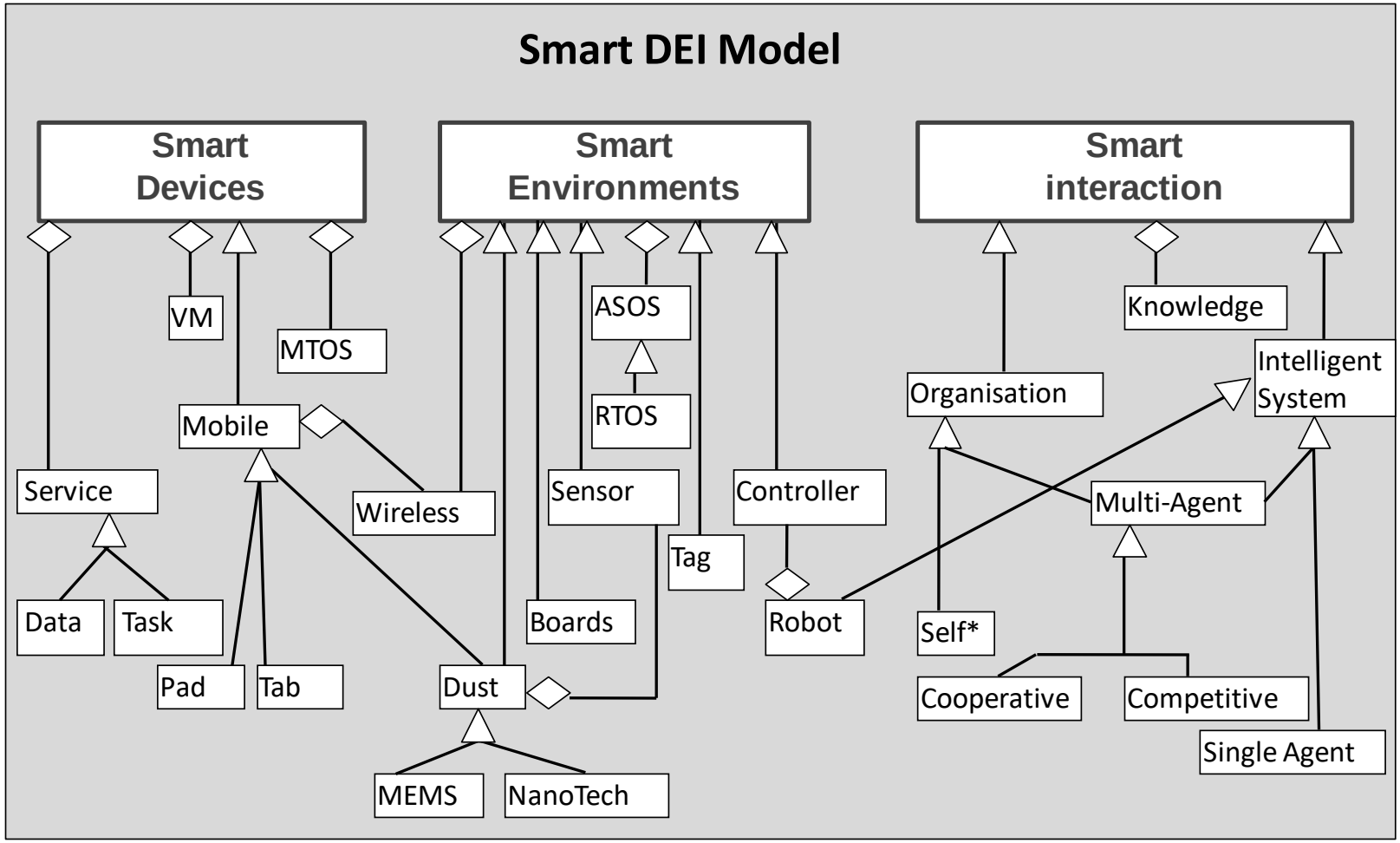
UbiCom System Model: Smart DEI



Smart DEI Model: types of smart device

- 3 main types of UbiCom system may themselves contain smart sub-systems at a lower level of granularity
  - e.g., a smart environment device may consist of smart sensors and smart controllers

UbiCom System: Smart Sub-Systems or Components



Smart Devices

- The main characteristics of smart devices are as follows: mobility, dynamic service discovery and intermittent resource access (concurrency, upgrading, etc.).

- There are various different forms for smart devices:
  - ICT Device Forms: Tabs, Pads and Boards
  - Extended Forms for ICT Devices: Dust, Skin and Clay
  - Mobility

ICT Device Forms: Tabs, Pads and Boards

- Devices tend to become smaller and lighter in weight, cheaper to produce.
- Devices can become prevalent, made more portable and can appear less obtrusive.

Weiser proposed a range of device sizes

1. *Smart Tabs*
2. *Smart Pads*:
3. *Smart Boards*:

Extended Forms for ICT Devices: Dust, Skin and Clay

Form Factors can be extended to support

1. **Smart Dust** *:* ICT devices can be miniaturised without visual output displays, e.g., Micro Electro Mechanical Systems (MEMS), ranging from nanometres through micrometres to millimetres. This form is called Smart Dust.

2. **Smart Skins :** Second, fabrics based upon light emitting and conductive polymers, organic computer devices, can be formed into more flexible non planar display surfaces and products such as clothes and curtains. This form is called Smart Skins.

3. **Smart Clay :** ensembles of MEMS can be formed into arbitrary three dimensional shapes as artefacts resembling many different kinds of physical object. This form is called Smart Clay

Mobility

Mobile devices usually refer to communicators, multimedia entertainment and business processing devices designed to be transported by their human owners, e.g., mobile phone, games consoles, etc.

There is a range of different types of mobiles as follows:

- Accompanied
- Portable
- Hand held
- Wearable
- Implanted or embedded

Smart Environment (Devices)

- In a smart environment, computation is seamlessly used to enhance ordinary activities.

- smart environment is ‘one that is able to acquire and apply knowledge about the environment and its inhabitants in order to improve their experience in that environment’.

- A smart environment consists of a set of networked devices that have some connection to the physical world.

Types of Smart Environment Device Interaction

- Tagging and Annotating
- Sensing & monitoring
- Filtering
- Adapting
- Controlling
  - Assembling
  - Regulating

Smart Interaction

- Smart interaction is needed to promote a unified and continuous interaction model between UbiCom applications and their UbiCom infrastructure, physical world and human environments.

- Smart interaction design model, system components dynamically organise and interact to achieve shared goals.

- A range of levels of interaction between UbiCom system components exists from basic to smart.

- Basic interaction that uses fixed interaction protocols between two statically linked dependent parties, whereas smart interaction that uses richer interaction protocols between multiple dynamic independent parties or entities.

Basic Interaction

- Typically involves two interlinked parties, a sender and a receiver.
- Sender knows things in advance.
- Two main types of basic interaction synchronous versus asynchronous

Smart Interaction

Smart Interaction extends basic interactions as follows.

- Coordinated interactions
- Cooperative versus competitive interaction
- Policy and Convention based Interaction
- Dynamic Organisational Interaction
- Semantic and Linguistic Interaction

Smart DEI Model Summary

- Basic Smart Device has many variations
  - 6 different physical form factors
  - 5 different groups of internal properties & over 70 sub-properties
- Multiple flavours of smart device,
  - e.g., Smart Mobile type of Smart device
  - e.g., Smart Environment type of Smart Device
- UbiCom System interact across 3 main types of environment: physical, virtual & human
- System of systems models in terms of multiple device combinations and interactions

Smart devices and services

Smart devices and service characteristics

- Smart devices are characterised by the

✓ ability to execute multiple,

✓ possibly concurrent,

✓ applications,

✓ mobility and customisation

✓ remote service access

✓ operating according to local resource constraints.

Service architecture model

it consists of

- partitioning and distribution of service components.
- Multi-tier client server models.
- Middleware
- Service oriented computing
- Grid computing
- Peer to Peer system

Partitioning and distribution of service components

- Range of designs for dividing and distributing services that depend on:

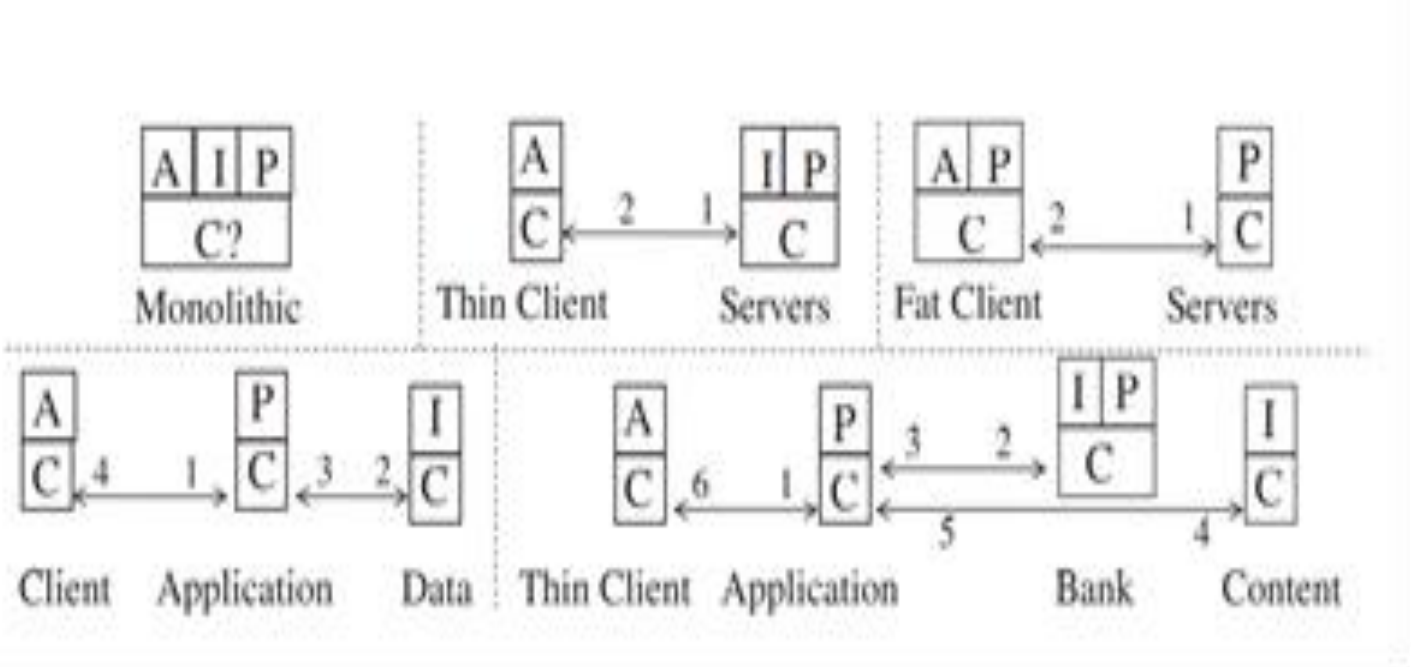
(1) the application

(2) the type of communication service

(3) the type of access device used.

- There are two types of resource devices high source devices and low resource devices.
- Consider the example of chess application, two different ways in which a chess application can be designed to run on smart devices.
- If we consider the example of chess application In a high resource device and in low resource devices.

Multi-tier client service models



- **In a single tier**

The whole application service resides locally, when it is operating. The system may be networked so that under special conditions it can go online to seek help when its operation is interrupted or because of local failures.

- **In a two tier**

Thin client server, the access device (or client device or terminal) supports data access or presentation, service processes execute remotely and the information associated with these services is stored remotely.

- **In multi-tier (3, 4 ... N tier) systems**

Rather than access devices being directly connected to the end service nodes, different numbers of intermediate nodes can be used.

It includes three parts

- **Distributed data storage**

  It includes : Transition monitor

  Centralised analysis of centrally stored data subset

  Mapping of a database

- **Distributed processing**

Divide the processing, distribute it among multiple remote processors, each executive part of the processing in parallel and then reassemble the result from the individual pieces to form the whole

- **Client server design**

The client server model has the advantage of more centralised control of distribution. The client server model is an asymmetric distributed computing model with respect to the resources and the direction of the interaction.

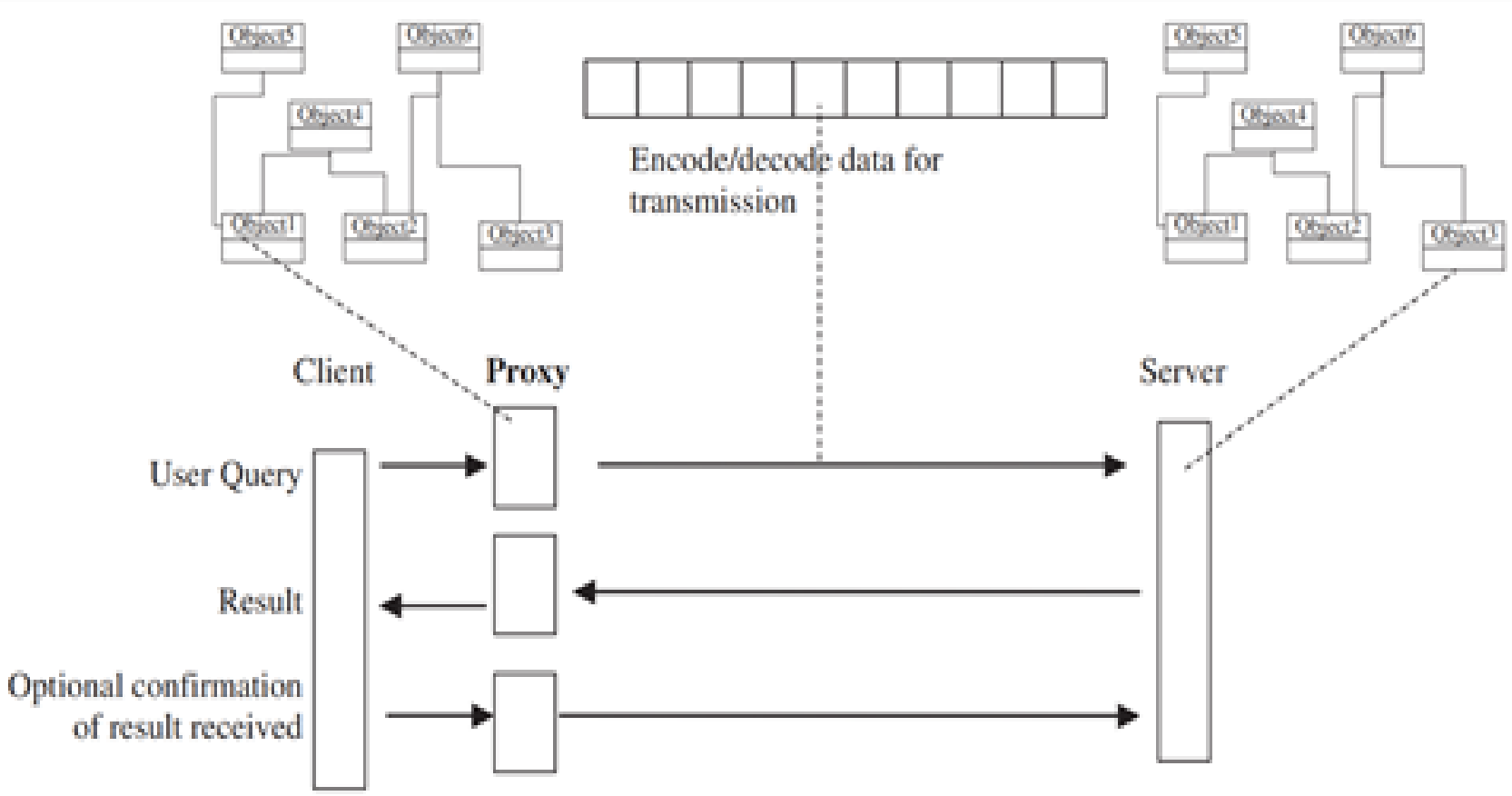
- **Proxy based service access**

Some applications, in order to mask the complexity of communication from being supported in client access process, use a client proxy.

Proxy based service access can:

- offload presentation processing and network processing
- hide the heterogeneity
- simplify and compose access to multiple service providers.
- reduce the complexity of communication used in access devices
- enable devices to operate intermittently in a disconnected state.
- shield network based applications from the mobility of the access devices

- **Proxy based service access**



Middleware

- Middleware was introduced in between applications and the operating system, to enable applications to hide and simplify access to the heterogeneous and distributed resources of multiple networked computing systems.

- Middleware essentially factors out a set of generic services, e.g., database access, file system access, messaging, time service, directory service, etc., out of the application services and out of the operating so that they can be application and operating system independent.

Service Oriented Computing (SOC)

- SOA is ‘a paradigm for organizing and utilizing distributed capabilities that may be under the control of different ownership domains’.

- service can be characterised in terms of:
  - Descriptions
  - Outcomes
  - Offers
  - Competency
  - Execution
  - Composition
  - Constraints or policies

Services in a SOA can be separated into three layers of functions:

- Basic (lower)
- Composition (middle)
- Management (higher layer).

These functions are described as follows:

- Service discovery (Basic function)
- Enterprise service bus (Basic function)
- Service invocation (Basic function)
- Service composition
- Service management

Grid Computing

- Grid computing refers to distributed systems that enable use and sharing of distributed resources.
- Three main types of grid system occur in practice:

(1) Computational grids

(2) Data grids

(3) Service grids :

Demand

Collaborative

Multimedia grid systems

- Grids focus on providing a single virtual computer view of distributed systems made up of heavyweight servers and fat client computers that communicate on high bandwidth highly available fixed networks.

Peer-to-Peer Systems

- They can be defined as distributed systems consisting of interconnected nodes able to self organise into network topologies with the purpose of sharing resources.

- A P2P computing model offers several important benefits:

➢ Lower cost of ownership for content sharing

➢ Performance enhancements

➢ Resource utilisation and sharing

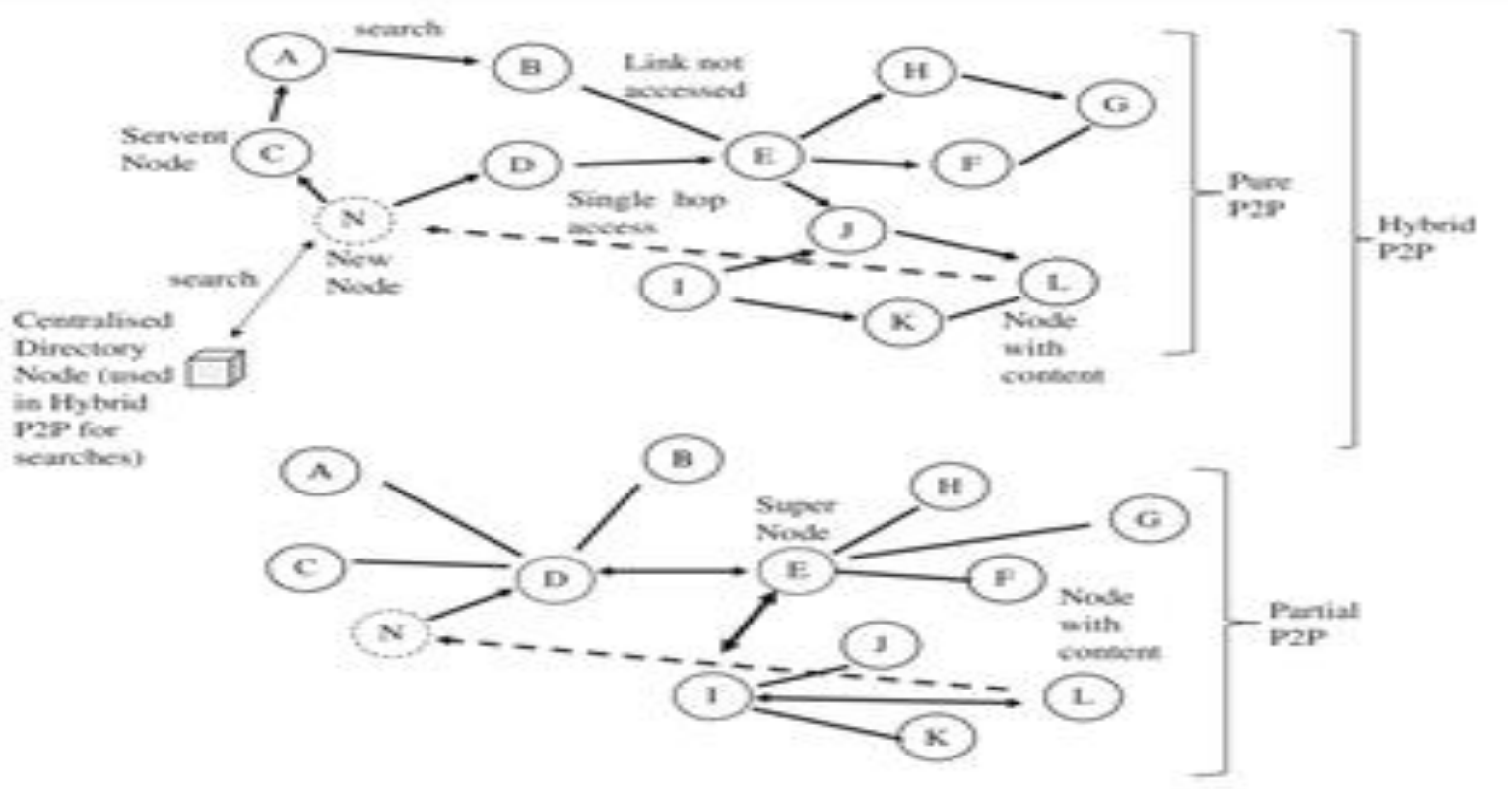
➢ Fault tolerance

- There are three main variations of P2P system depending on the types of computer nodes:

➢ Pure, decentralised, P2P

➢ Partially decentralised (SuperNode) P2P

➢ Hybrid decentralised P2P



Service Provision Life-Cycle

- The design of service provision must contend with intermittent service access and handovers between different service instances, e.g., wireless communication handovers.

- There are two separate aspects to this:

First, defining a generic life cycle model for service provision

Second , to manage this life cycle.

Service Provision Life-Cycle

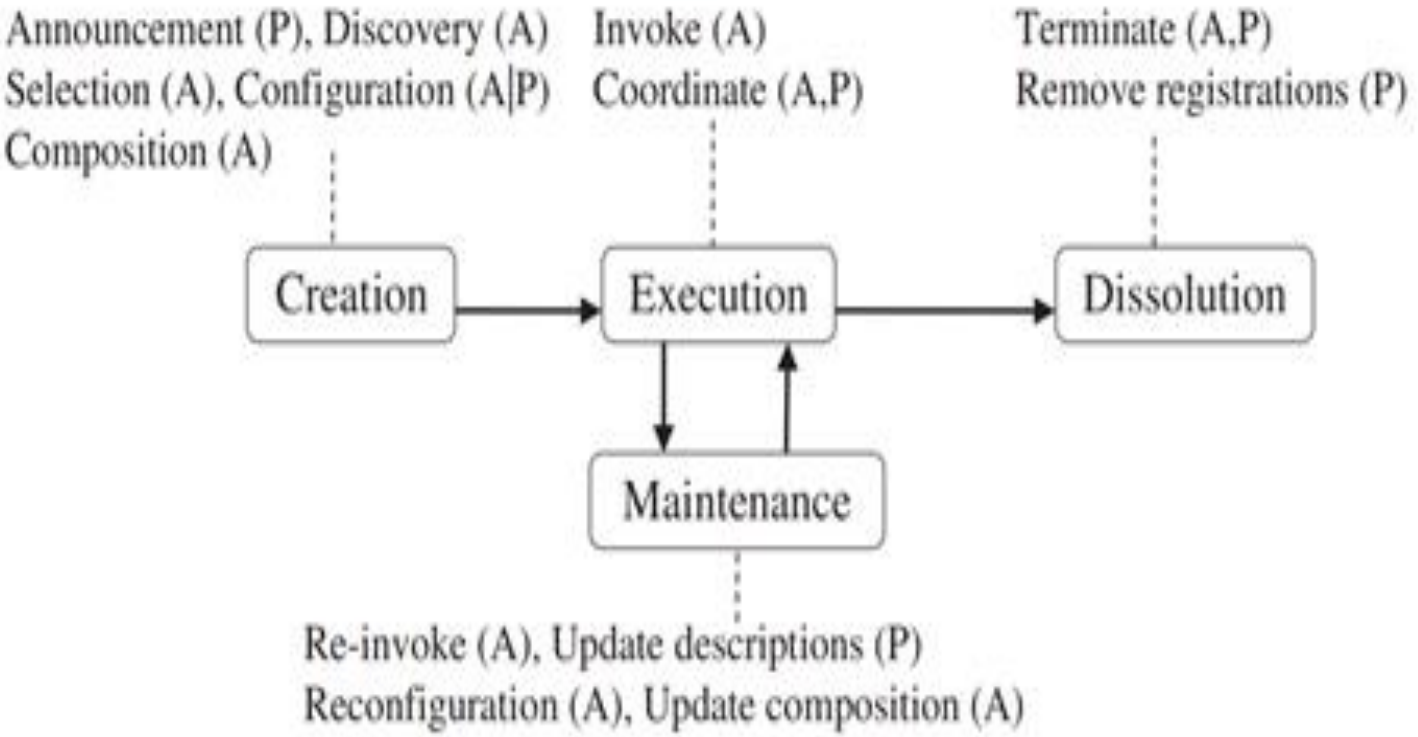


Figure 3.9 The service life cycle: smart services entail operation and management throughout the whole life cycle. P and A indicate that service processes and service access are active during each phase

- In a simple service provision lifecycle model, only two of the five service model components are active, the processing services or service provision and service access or clients, the other three components, communication, stored information and information sources, are treated as passive components.

- We have a we use different phases in the service provision life cycle:

✓ service creation phase

✓ operational or service execution phase

✓ service maintenance phase

✓ service dissolution phase

- Different things have been explained for service provision life cycle:

- Network Discovery
- Service Announcement, Discovery, Selection and Configuration
- Service Invocation
- Service Composition

Network Discovery

- Network discovery must precede service registration and service discovery.
- Dynamic network discovery is used by mobile nodes and when new nodes are introduced into a network.
- A Domain Name Service or DNS is used to map an IP address to a name of some network node and vice versa.
- A common approach to discover the network is to use DHCP (Dynamic Host Control Protocol) to ask a DHCP server for an IP address that is leased for a given time.

Service Announcement, Discovery, Selection and Configuration

- If service providers and requesters are static, then there is little need for dynamic service discovery.

- Dynamic service discovery is needed to allow
  - service requester to change provider when requester or provider are mobile.
  - When network exists is intermittent.
  - When requestor or provider fails.

There are two main approaches for dynamic service announcement and discovery

- Pull
- Push

Pull

pull use is broadcast or multicast the available service request or service capabilities to all the unknown parties example Bluetooth.

Push

Push uses look up to search or browse list of request or capabilities previously announced two or discovery held by some known third party.

Service selection

There are several dimensions for service selection

- Request base versus capability based versus gold based.
- Exact versus inexact.
- Syntactical versus semantic.

Service invocation

Two possibilities are there

- Requester need to know how to invoke the service.
- Requester may not know in which order to invoke service action or how to handle message sequence in a process without terminating service process

The mean service invocation design include

- Distributed processes
- Asynchronous versus synchronous communication model
- Reliable versus unreliable communication
- On demand service access
- Event driven architecture
- Shared data repository
- Enterprise service bus model
- Volatile service invocation

Service composition

- Composition is concerned with synthesising new services and assembling more complex (composite) services from simple (atomic) services to achieve a user or application goal, and then collectively executing them as composite service processes.

- Service processes are sequences of individual service actions that are scheduled for execution.

- Service processes may involve one or more entities, one or more actions and involve one or more processes.

Applications and requirements

Smart Devices: CCI

- **Smart Boards, Pads and Tabs**

A large wall display program called LiveBoard which migrated from amorphous silicon to rear screen projection; smaller computers, the book sized MPad and the palm sized ParcTab computer.

- **Active Badge, Bat and Floor**

First context aware computing application, designed to enhance user mobility and to support location awareness.

Active Bat, based on ultrasound4 that could locate people up to an accuracy of 3 cm.

The active floor or smart floor did not require someone to explicitly carry some identifying token. Instead, this type of device is designed to identify people indirectly, e.g., in this case by their type of walk or gait

Smart Environments: CPI and CCI

- **Classroom 2000**

- Classroom 2000 in an attempt to support both teaching and learning.

- Microphones and video cameras were embedded in the ceiling and the signals from the microphones and cameras were stored. The electronic whiteboard was used again but this time networked. Two ceiling mounted projectors attached to networked computers were also for display.

- **Smart Space and Meeting Room**

- Smart Space and Meeting Room Projects, focused on the use of pervasive devices, sensors, and networks to provide an infrastructure for context aware smart meeting rooms that sense ongoing human activities and respond to them.

- The Meeting Room digitised signals from two hundred microphones, five video cameras, i.e. direct sensor data, and had a smart whiteboard.

- **Interactive Workspaces and iRoom**

- Interactive rooms called iRooms that contained one or more large displays, with the ability to integrate portable devices and to create applications integrating the use of multiple devices in the user space.

- The second version of the iRoom contained three touch sensitive whiteboard sized displays along the side wall, a large display supporting pen interaction called the interactive mural built into the front wall and a custom designed display to look like a standard conference room table.

- The room also had cameras, microphones, wireless LAN support, and a variety of wireless buttons and other interaction devices. The room supported three main interactive tasks: moving data, moving control and dynamic coordination of multiple applications.

- **Cooltown**

- Support mobile users providing access to information via wireless hand held devices based upon Web technology and linking the virtual ICT world to the physical world.

- A key feature of the Cooltown approach is that a physical world resource can have a Web presence.

- Physical resources are associated with a simple standard resource identifier, a URL or Universal Resource Locator.

- **EasyLiving and SPOT**

- The focus of the EasyLiving was on developing intelligent environ ments which allow the dynamic aggregation of diverse I/O devices into a single coherent user experience.

- In order to allow content created in one device to be output to a different device and controlled by yet another device, or because the input control device is not conveniently collocated with a remote display device, such as when remotely controlling a large screen display.

- EasyLiving enables this kind of flexible interac tion by providing abstract descriptions of their capabilities, geometric modelling of the location of devices in relation to other devices and through sensing capabilities.

- In the SPOT initiative, SPOT devices were designed to listen for digitally encoded data such as news stories, weather forecasts, personal messages, traffic updates, and retail directories trans mitted on frequency sidebands leased from commercial FM radio stations